

BGT60TR13C

60 GHz Radar Sensor

Datasheet V2.4.6

Features

- 60 GHz radar sensor for FMCW operation
- 5.5 GHz bandwidth
- Antenna-in-package (6.5 x 5.0 x 0.9 mm³)
- Digital interface for chip configuration and radar data acquisition
- Optimized power modes for low-power operation
- Integrated state machine for independent operation

Potential applications

- Radar frontend for gesture sensing
- High resolution FMCW radars
- Short range sensing operations
- Hidden sensing applications behind radome

Product validation

Packaged device qualified according to JEDEC 20/22.

Description

The BGT60TR13C, a 60 GHz radar sensor with antenna in package, enables ultra-wide bandwidth FMCW operation in a small package. Sensor configuration and data acquisition are enabled with a digital interface and the integrated state machine enables independent data acquisition with power mode optimization for lowest power consumption.

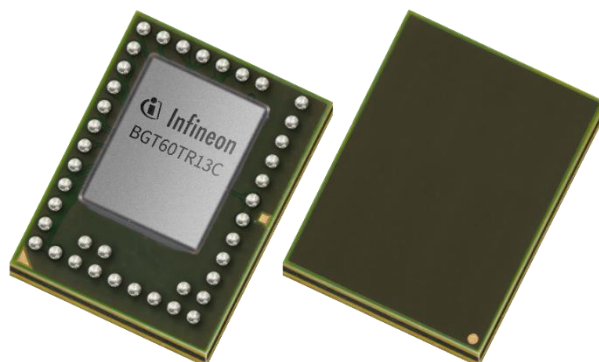


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Introduction

1 Introduction

New smart sensors for gesture recognition can be based on radar systems, in special case, FMCW radars. Those systems can comprise several blocks: Radio Frequency (RF) front-end, Analog Base Band (ABB), Analog to Digital Converter (ADC), Phase Locked Loop (PLL), memory (FIFO e.g.), Serial Peripheral Interface (SPI) and Antennas. Smart sensors require a high level of integration, thus, the components listed above should be integrated in a single chip solution. BGT60TR13C offers this level of integration in a single chipset.

1.1 Product Overview

The core functionality of BGT60TR13C is to transmit frequency modulated continuous wave (FMCW) signal via one of the transmitter channel (TX) and receive the echo signals from the target object on the three receiving channels (RX). Each receiver path includes a baseband filtering, a VGA, as well as an ADC. The digitized output is stored in a FIFO. The data are transferred to an external host, microcontroller unit (MCU) or application processor (AP), to run radar signal processing. A typical implementation of a sensor system consists of two main blocks only (see Figure 1):

- BGT60TR13C handles the RF signals and provides the sampled IF signals
- Application Processor which captures and processes the radar signals

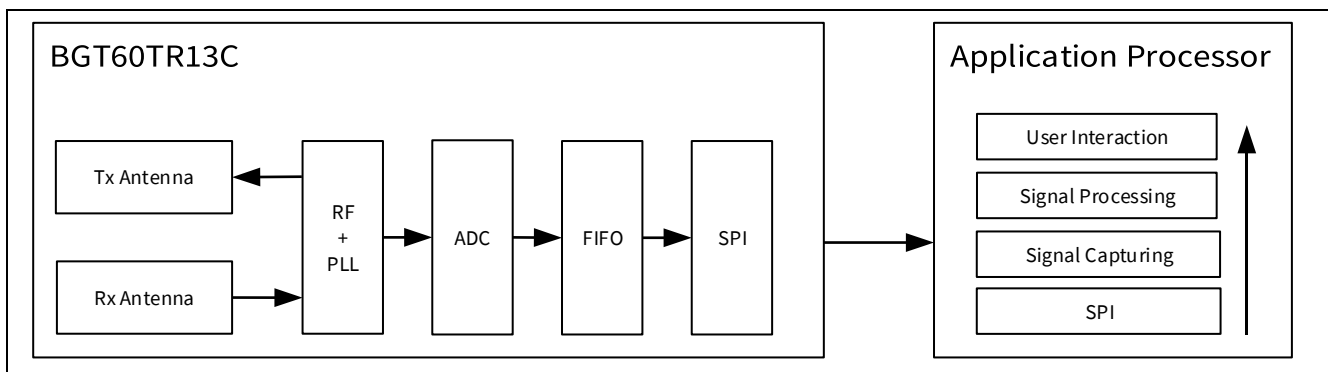


Figure 1 Data flow in the complete radar sensor system

1.2 Potential Applications

The chipset has been designed to address mainly the following potential applications:

- Radar frontend for gesture sensing
- High resolution FMCW radars
- Short range sensing operations
- Hidden sensing applications behind radome

Introduction

1.3 BGT60TR13C Bare Die Block Diagram

BGT60TR13C block diagram is presented in Figure 2.

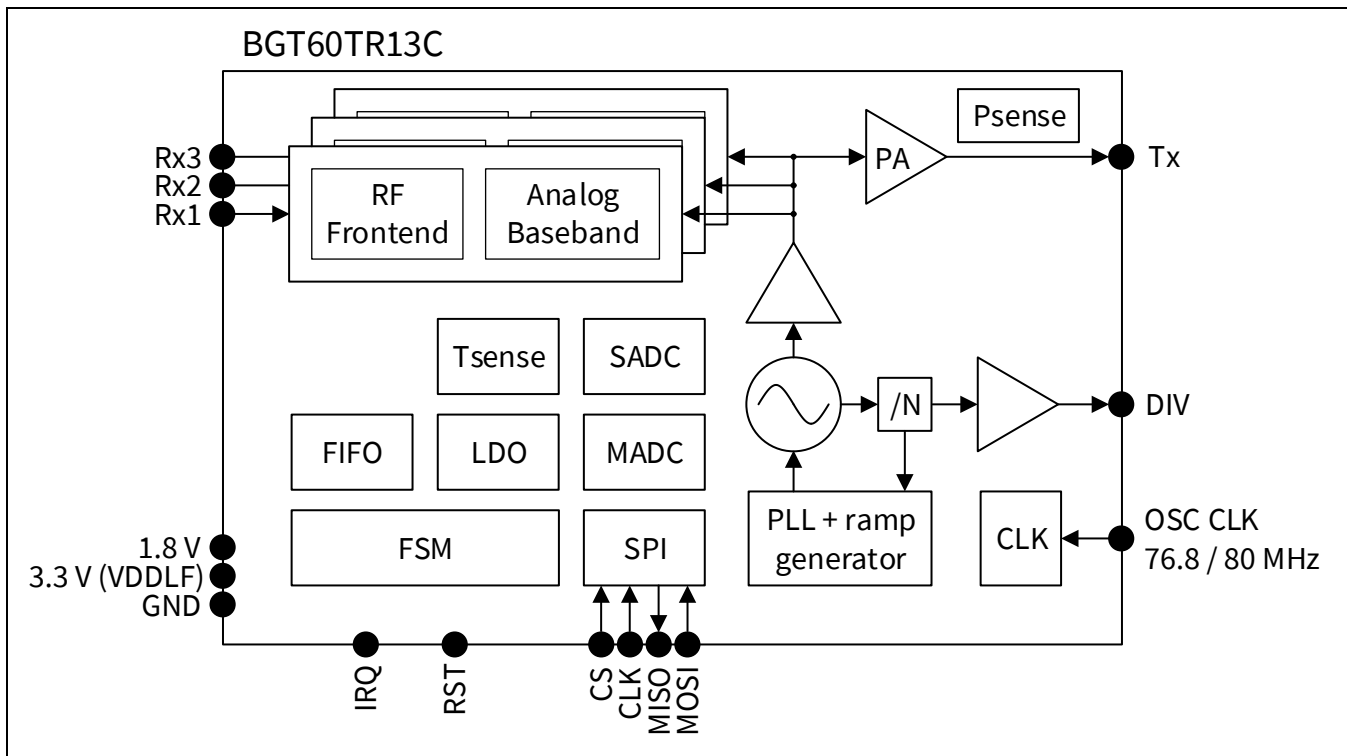


Figure 2 BGT60TR13C Bare die block diagram

Feature List:

- Single supply voltage level of 1.8 V for both, digital and analog domains
- Integrated LDOs from 1.8 V to 1.5 V to supply the digital domain
- RF-Frontend at 60 GHz covering frequencies from 58.0 to 63.5 GHz with one TX and three RX channels
- Baseband chain consisting of high pass filter, low noise voltage gain amplifier (VGA), and antialiasing filters
- Three ADC channels with 12 bits resolution and up to 4 MSps sampling rate to sample the RX-IF channels
- Integrated RF-PLL, timers, counters, and FSM to run set of frames in standalone mode (no communication with AP required except first trigger and raw data transfer)
- Full duplex FIFO structure as data buffer (196 kbit = 8192 words x 24 bits)
- Linear Feedback Shift Register (LFSR) test pattern generator on chip for data transfer check
- 8 to 10 bits sensor ADC for power and temperature measurement
- Standard SPI mode for configuration and status register read accesses
- Dedicated power modes for power reduction
- An external 80 MHz reference oscillator is used as a system clock source
- BITE (Built in test equipment) for EOL test in production at Infineon to verify RF performance
- Fabricated with BiCMOS Infineon process technology
- Housed in a laminate package
- Antennas integrated in the redistribution layers of the package

Introduction

Table 1 Ball and Antenna Definition

Ball	Function
Rx3	Receiver ch3

1.4.1 IO and Supply Pins

The following Table 2 gives an overview on the input/output pins of BGT60TR13C.

Table 2 BGT60TR13C Input/Output Pins

Symbol	Type	Domain	Description	Domain
DIV_TEST	A _{OUT}	VDDRF	VCO divided by 16 output	Analog-RF
OSC_CLK	A _{IN}	VDDRF	80 MHz (e.g.) Xtal input	Analog-RF
CLK	D _{IN}	VDDD	SPI CLK input	SPI
IRQ	D _{OUT}	VDDD	Interrupt output	Control FSM
CS_N	D _{IN}	VDDD	SPI chip select input, active low	SPI
DI	D _{IN}	VDDD	SPI signal from the host output	SPI
DO	D _{OUT}	VDDD	SPI signal to the host input	SPI
DIO3	D _{IN} / D _{OUT}	VDDD	HW reset pin	SPI

The power supply pins are described in Table 3.

Table 3 BGT60TR13C Supply Pins

Symbol	Type	Domain	Description	Domain
VDDD	V _{IN}	1.8 V	Digital supply voltage	Digital
VDDA	V _{IN}	1.8 V	Analog supply voltage	ADC
VAREF	V _{OUT}	1.2 V	Positive reference voltage output; for bypass cap	ADC
VDDVCO	V _{IN}	1.8 V	Analog supply voltage to the VCO	Analog-RF
VDDRF	V _{IN}	1.8 V	Analog supply voltage	Analog-RF
VDDL	V _{IN}	3.3V	Analog supply voltage for the level shifter for the PLL loop filter	Analog-RF
VDDPLL	V _{IN}	1.8 V	Analog supply voltage to the PLL	Analog-RF
VSSRF	GNDA	0 V	Analog ground connection	Analog-RF
VSSA	GNDA	0 V	Analog ground connection	ADC
VSSD	GNDD	0 V	Digital ground connection	Digital

Abbreviations:

V_{IN} ... supply voltage input pin

D_{IN} ... digital input pin

D_{OUT} ... digital output pin

V_{OUT} ... supply voltage output pin

A_{OUT} ... analog output pin

Introduction

A_{IN} ... analog input pin

GNDA ... analog ground connection

GNDD ... digital ground connection

1.5 BGT60TR13C Functional Block Diagram

BGT60TR13C consists of some main functional blocks:

- **Antenna** built in package, see Figure 31
- **RF Frontend** consisting of 3ch Rx, 1ch Tx, LO generation, and divider by 4/5
- **ABB**, analog baseband consisting of high pass filter (HPF), VGA, anti-aliasing filter (AAF)
- **PLL**, 3rd order sigma-delta based to perform FMCW ramp
- **MADC**, 3ch 12 bits differential SAR ADCs interfaced to the ABB via a driver and to the FIFO via a mux, see paragraph **Error! Reference source not found.**
- **SADC**, 8 to 10 bits single-ended SAR ADC used to sense the sensor data
- **FIFO**, 196 kbit= 8192 words x 24 bits
- **Register banks**, 127 registers, see paragraph 3
- **SPI**, up to 50 MHz clock in standard mode
- **FSM**, finite state machine which manage the complete chip
- Clock wise, two domains can be identified:
 - 80 MHz system clock (SYS_CLK) domain for PLL, MADC, SADC, and FIFO
 - 50 MHz (e.g.) SPI clock

The main FSM syncs those two domains.

Introduction

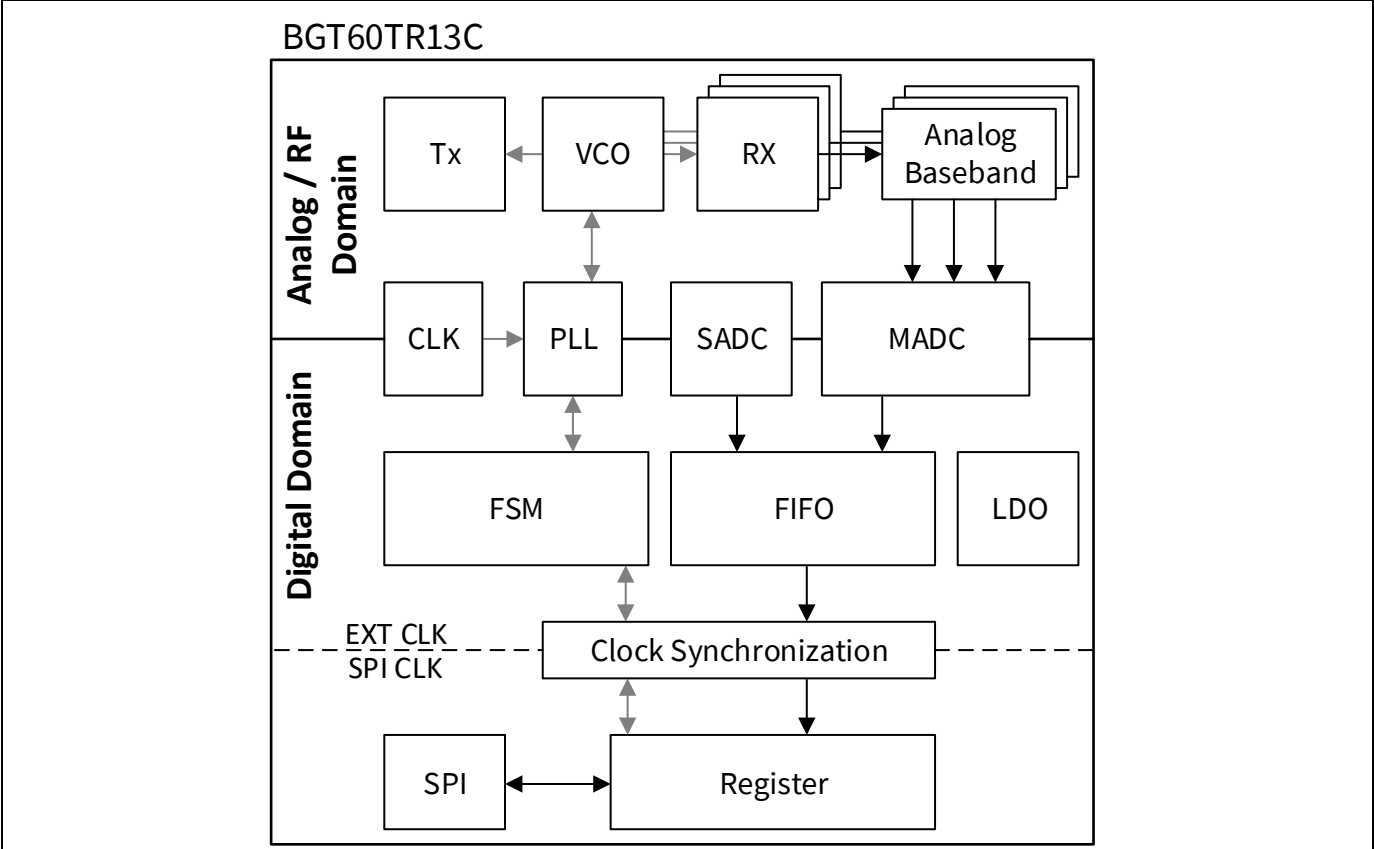


Figure 4 **BGT60TR13C functional overview**

General Product Specification

2 General Product Specification

The reference for all specified data is the Infineon application board, available on request.

2.1 Absolute Maximum Ratings

Table 4 Absolute Maximum Ratings $T_b = -40^{\circ}\text{C}$ to 105°C , all voltages with respect to ground, positive current flowing into pin (unless otherwise specified). Parameters not subject to production test

Spec	Symbol	Unit	Value			Condition
Parameter			Min	Typ	Max	
Supply Voltage	VDDD	V	-0.3		+2	
Supply Voltage	VDDA	V	-0.3		+2	
Supply Voltage	VDDRF	V	-0.3		+2	
Supply Voltage	VDDVCO	V	-0.3		+2	
Supply Voltage	VDDPLL	V	-0.3		+2	
Supply Voltage	VDDLf	V	-0.3		+3.7	
DC Voltage at all I/O Pins	$V_{I/O}$	V	-0.3		VDD+0.3	Not exceeding 2V
RF Input Power Level	PRF	dBm			+10	At the Rx input-port
Junction Temperature	T_j	$^{\circ}\text{C}$	-40		+125	
Storage Temperature	T_{stg}	$^{\circ}\text{C}$	-40		+150	

Warning: Stresses above the maximum values listed here may cause permanent damage to the device. Maximum ratings are absolute ratings; exceeding only one of these values may cause irreversible damage to the integrated circuit. Exposure to conditions at or below absolute maximum rating but above the specified maximum operation conditions may affect device reliability and lifetime. Functionality of the device might not be given under these conditions.

2.2 Range of Functionality

Table 5 Range of Functionality, VDDD= 1.71 to 1.89 V, $T_b = -20$ to $+70^{\circ}\text{C}$

Spec	Symbol	Unit	Value			Condition
Parameter			Min	Typ	Max	
Supply Voltage	VDDD	V	1.71	1.8	1.89	Noise on each supply domain should not exceed the level of 20 μVpp in the frequency range 20kHz-700kHz ⁽¹⁾
Supply Voltage	VDDA	V	1.71	1.8	1.89	
Supply Voltage	VDDRF	V	1.71	1.8	1.89	
Supply Voltage	VDDVCO	V	1.71	1.8	1.89	
Supply Voltage	VDDPLL	V	1.71	1.8	1.89	
Supply Voltage	VDDLf	V	2.5	3.3	3.63	
Chip Backside Temperature	T_b	$^{\circ}\text{C}$	-20		70	Measured with the on chip temperature sensor
Frequency Range	f_{RF}	GHz	58.0		63.5	

General Product Specification

System Reference Frequency	$f_{\text{SYS_CLK}}$	MHz	75	80	85	1.8 V CMOS clock; 78 MHz not allowed
Duty cycle of $f_{\text{SYS_CLK}}$	f_{DUTSYS}	%	45	50	55	
Rise and Fall Time of $f_{\text{SYS_CLK}}$	$t_{\text{RS,FS,SYS}}$	ns			6	
Phase Jitter of $f_{\text{SYS_CLK}}$	J_{PHYSYS}	ps		1		BW: 12kHz to 20MHz

- ¹⁾ This value will guarantee no artifact/false target in the Range-Doppler map when it is calculated with a minimum of 8 chirps.

2.3 Current Consumption

Table 6 Overall Current Consumption, VDD (all except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
Parameter			Min	Typ	Max	
Idd Deep Sleep ¹⁾	$I_{\text{dd}_{\text{ds}}}$	mA	0.05	0.12	0.555 ⁵⁾	
Idd Idle ²⁾	$I_{\text{dd}_{\text{idle}}}$	mA	0.05	2.8	5	
Idd Init0, 3Rx+ 1Tx	$I_{\text{dd}_{\text{int0}}}$	mA	136	175	205	
Idd Init1, 3Rx+ 1Tx ³⁾	$I_{\text{dd}_{\text{int1}}}$	mA	141	185	215	
Idd Active, 3Rx + 1Tx ⁴⁾	$I_{\text{dd}_{\text{act}}}$	mA	160	201	230	

- ¹⁾ All registers in reset mode, 80 MHz clock path disabled
- ²⁾ MADC band-gap running
- ³⁾ Idd for the rest of interchirp similar to Init1
- ⁴⁾ Device set in radar mode, DAC Tx set to #31
- ⁵⁾ The value at max refers to the max temperature, +70°C, and the max supply, 1.89 V

Table 7 VDDD Domain Current Consumption, VDD (any except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
Parameter			Min	Typ	Max	
Idd Deep Sleep ¹⁾	$D I_{\text{dd}_{\text{ds}}}$	mA	0.05	0.1	0.48 ⁵⁾	
Idd Idle ²⁾	$D I_{\text{dd}_{\text{idle}}}$	mA	1.5	2.5	3.5	
Idd Init0, 3Rx+ 1Tx	$D I_{\text{dd}_{\text{int0}}}$	mA	2	3	4	
Idd Init1, 3Rx+ 1Tx ³⁾	$D I_{\text{dd}_{\text{int1}}}$	mA	3	4	5	
Idd Active, 3Rx + 1Tx ⁴⁾	$D I_{\text{dd}_{\text{act}}}$	mA	3	4	5	

- ¹⁾ All registers in reset mode, 80 MHz clock path disabled
- ²⁾ MADC band-gap running
- ³⁾ Idd for the rest of interchirp similar to Init1
- ⁴⁾ Device set in radar mode, FIFO in low power mode, DAC Tx set to #31 for an output power of +5dBm
- ⁵⁾ The value at max refers to the max temperature, +70°C, and the max supply, 1.89 V

General Product Specification

Table 8 VDDA Domain Current Consumption, VDD (any except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Parameter						
Idd Deep Sleep ¹⁾	AIdd _{ds}	mA	0	0.005	0.01	
Idd Idle ²⁾	AIdd _{idle}	mA	0.01	0.2	0.3	
Idd Init0, 3Rx+ 1Tx	AIdd _{int0}	mA	0.5	1.6	3	
Idd Init1, 3Rx+ 1Tx ³⁾	AIdd _{int1}	mA	0.5	1.6	3	
Idd Active, 3Rx + 1Tx ⁴⁾	AIdd _{act}	mA	0.5	1.6	3	

¹⁾ All registers in reset mode, 80 MHz clock path disabled

²⁾ MADC band-gap running

³⁾ Idd for the rest of interchirp similar to Init1

⁴⁾ Device set in radar mode, DAC Tx set to #31

Table 9 VDDPLL Domain Current Consumption, VDD (any except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Parameter						
Idd Deep Sleep ¹⁾	PLLId _{ds}	mA	0	0.005	0.01	
Idd Idle ²⁾	PLLId _{idle}	mA	0	0.005	0.01	
Idd Init0, 3Rx+ 1Tx	PLLId _{int0}	mA	0.6	0.9	1.2	
Idd Init1, 3Rx+ 1Tx ³⁾	PLLId _{int1}	mA	5	8	10	
Idd Active, 3Rx + 1Tx ⁴⁾	PLLId _{act}	mA	5	8	10	

¹⁾ All registers in reset mode, 80 MHz clock path disabled

²⁾ MADC band-gap running

³⁾ Idd for the rest of interchirp similar to Init1

⁴⁾ Device set in radar mode, DAC Tx set to #31

Table 10 VDDL F Domain Current Consumption, VDD (any except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Parameter						
Idd Deep Sleep ¹⁾	LFId _{ds}	mA	0	0.005	0.01	
Idd Idle ²⁾	LFId _{idle}	mA	0	0.005	0.01	
Idd Init0, 3Rx+ 1Tx	LFId _{int0}	mA	0.2	0.45	0.5	
Idd Init1, 3Rx+ 1Tx ³⁾	LFId _{int1}	mA	0.2	0.45	0.5	
Idd Active, 3Rx + 1Tx ⁴⁾	LFId _{act}	mA	0.2	0.45	0.5	

¹⁾ All registers in reset mode, 80 MHz clock path disabled

²⁾ MADC band-gap running

General Product Specification

- ³⁾ Idd for the rest of interchirp similar to Init1
- ⁴⁾ Device set in radar mode, DAC Tx set to #31

Table 11 VDDRF + VDDVCO Domain Current Consumption, VDD (any except LF)= 1.71 to 1.89 and Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Idd Deep Sleep ¹⁾	RFIdd _{ds}	mA	0	0.02	0.045	
Idd Idle ²⁾	RFIdd _{idle}	mA	0	0.02	0.045	
Idd Init0, 3Rx+ 1Tx	RFIdd _{int0}	mA	133	170	196	
Idd Init1, 3Rx+ 1Tx ³⁾	RFIdd _{int1}	mA	133	170	196	
Idd Active, 3Rx + 1Tx ⁴⁾	RFIdd _{act}	mA	151	187	212	

- ¹⁾ All registers in reset mode, 80 MHz clock path disabled
- ²⁾ MADC band-gap running
- ³⁾ Idd for the rest of interchirp similar to Init1
- ⁴⁾ Device set in radar mode, DAC Tx set to #31

2.4 ESD Integrity

Table 12 ESD Integrity, VDD(any)= 1.71 to 1.89 V, Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
ESD robustness, HBM All pins	V _{ESD-HBM}	V	-2000		+2000	According to JS-001 (R = 1.5 kΩ, C = 100 pF)
ESD robustness, CDM All pins except M2	V _{ESD-CDM}	V	-500		+500	According to JS-002
ESD robustness, CDM Pin M2	V _{ESD-CDM, M2}	V	-250		+250	According to JS-002

CDM: Field-Induced Charged-Device Model ANSI/ESDA/JEDEC JS-002. Simulates charging/discharging events that occur in production equipment and processes. Potential for CDM ESD events occurs whenever there is metal-to-metal contact in manufacturing.

HBM: Human Body Model ANSI/ESDA/JEDEC JS-001 (R = 1.5 kΩ, C=100 pF).

General Product Specification

2.5 Thermal Resistance

Table 13 Thermal Resistance, VDD(any)= 1.71 to 1.89 V, Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
Parameter			Min	Typ	Max	
Package Rth	R _{th}	K/W		35		Chip backside to ambient temperature

2.6 Product Validation

Qualified for potential applications listed in section 1.2 based on the test conditions in the relevant tests of JEDEC20/22.

BGT60TR13C Registers

3 BGT60TR13C Registers

An array of registers visible via the SPI is used to control and program the states of the different blocks inside the chip.

3.1 Register List

The registers are arranged in blocks of 24 bits each. Each block is identified by its unique address. The registers are accessed from the SPI module. The bit fields from each register are arranged in MSB first order.

Table 14 The following table gives an overview on the BGT60TR13C registers. Register Overview / Address Table

Register Address	Register Name	Description		
0x00	MAIN	Main register		
0x01	ADC0	MADC control register		
0x02	CHIP_ID	Digital and RF version		
0x03	STAT1	Status register 1		
0x04	PACR1	PLL analog control register 1		
0x05	PACR2	PLL analog control register 2		
0x06	SFCTL	SPI and FIFO Control		
0x07	SADC_CTRL	Sensor ADC ctrl reg		
0x08	CSI_0	Channel set idle mode 0		
0x09	CSI_1	Channel set idle mode 1		
0x0A	CSI_2	Channel set idle mode 2		
0x0B	CSCI	Channel set control idle mode		
0x0C	CSDS_0	Channel set deep sleep mode 0		
0x0D	CSDS_1	Channel set deep sleep mode 1		
0x0E	CSDS_2	Channel set deep sleep mode 2		
0x0F	CSCDS	Channel set control deep sleep mode		
0x10	CSU1_0	Channel set 1 (up)		
0x11	CSU1_1	Channel set 1 (up)		
0x12	CSU1_2	Channel set 1 (up)		
0x13	CSD1_0	Channel set 1 (down)		
0x14	CSD1_1	Channel set 1 (down)		
0x15	CSD1_2	Channel set 1 (down)		
0x16	CSC1	Channel set control 1 (up/dn)		
0x17	CSU2_0	Channel set 2 (up)		
0x18	CSU2_1	Channel set 2 (up)		
0x19	CSU2_2	Channel set 2 (up)		
0x1A	CSD2_0	Channel set 2 (down)		
0x1B	CSD2_1	Channel set 2 (down)		
0x1C	CSD2_2	Channel set 2 (down)		

BGT60TR13C Registers

0x1D	CSC2	Channel set control 2 (up/dn)		
0x1E	CSU3_0	Channel set 3 (up)		
0x1F	CSU3_1	Channel set 3 (up)		
0x20	CSU3_2	Channel set 3 (up)		
0x21	CSD3_0	Channel set 3 (down)		
0x22	CSD3_1	Channel set 3 (down)		
0x23	CSD3_2	Channel set 3 (down)		
0x24	CSC3	Channel set control 3 (up/dn)		
0x25	CSU4_0	Channel set 4 (up)		
0x26	CSU4_1	Channel set 4 (up)		
0x27	CSU4_2	Channel set 4 (up)		
0x28	CSD4_0	Channel set 4 (down)		
0x29	CSD4_1	Channel set 4 (down)		
0x2A	CSD4_2	Channel set 4 (down)		
0x2B	CSC4	Channel set control 4 (up/dn)		
0x2C	CCR0	Chirp control register 0		
0x2D	CCR1	Chirp control register 1		
0x2E	CCR2	Chirp control register 2		
0x2F	CCR3	Chirp control register 3		
0x30	PLL1_0	FSU1 – shape 1		
0x31	PLL1_1	RSU1 – shape 1		
0x32	PLL1_2	RTU1 – Shape 1		
0x33	PLL1_3	AP1 – shape 1		
0x34	PLL1_4	FSD1 – shape 1		
0x35	PLL1_5	RSD1 – shape 1		
0x36	PLL1_6	RTD1 – shape 1		
0x37	PLL1_7	SCR – shape 1		
0x38	PLL2_0	FSU1 – shape 2		
0x39	PLL2_1	RSU1 – shape 2		
0x3A	PLL2_2	RTU1 – shape 2		
0x3B	PLL2_3	AP1 – shape 2		
0x3C	PLL2_4	FSD1 – shape 2		
0x3D	PLL2_5	RSD1 – shape 2		
0x3E	PLL2_6	RTD1 – shape 2		
0x3F	PLL2_7	SCR – shape 2		
0x40	PLL3_0	FSU1 – shape 3		
0x41	PLL3_1	RSU1 – shape 3		
0x42	PLL3_2	RTU1 – shape 3		
0x43	PLL3_3	AP1 – shape 3		
0x44	PLL3_4	FSD1 – shape 3		
0x45	PLL3_5	RSD1 – shape 3		

BGT60TR13C Registers

0x46	PLL3_6	RTD1 – shape 3		
0x47	PLL3_7	SCR – shape 3		
0x48	PLL4_0	FSU1 – shape 4		
0x49	PLL4_1	RSU1 – shape 4		
0x4A	PLL4_2	RTU1 – shape 4		
0x4B	PLL4_3	AP1 – shape 4		
0x4C	PLL4_4	FSD1 – shape 4		
0x4D	PLL4_5	RSD1 – shape 4		
0x4E	PLL4_6	RTD1 – shape 4		
0x4F	PLL4_7	SCR – shape 4		
0x55	RFT0	RF test register 0		
0x56	RFT1	RSVD		
0x59	PLL_DFT0	PLL DFT register 0		
0x5D	STAT0	Status register 0		
0x5E	SADC_RESULT	Sensor ADC result register		
0x5F	FSTAT	FIFO status register		
>= 0x60		FIFO access		

Note: Reserved bits (RSVD) in the registers should not be modified. They should be kept in the default/reset state unless otherwise specified.

3.1.1 Abbreviations

Access modes on the registers:

- R ... Readable register or bit field
- W ... Writeable register or bit field
- S ... Status bit can be set to readable mode “R”
- RSVD ... Reserved value which is not assigned at the moment

3.2 CHIP_ID

The register CHIP_ID provides information regarding the digital code version, the RF block version, and the antenna configuration (number of channels, position of the antennas e.g.).

It is used by the driver to configure the device properly according to the information above.

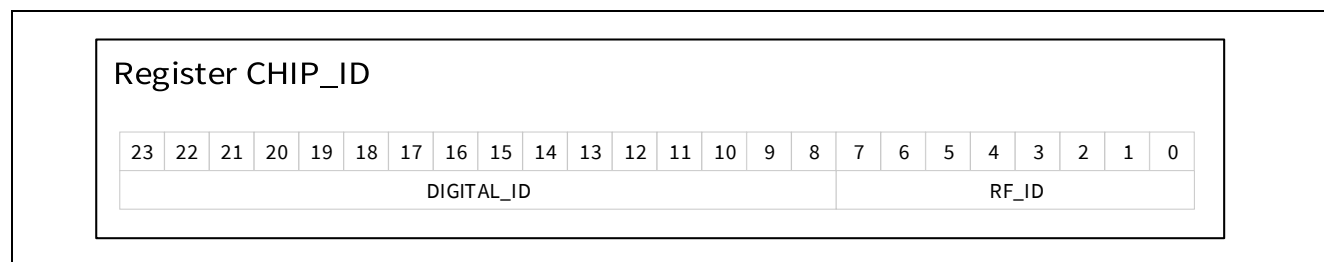


Figure 5 **CHIP_ID register**

BGT60TR13C Registers

Table 15 **CHIP_ID: Register Description**

Symbol	Bits	Type	Description	RST
DIGITAL_ID	23:8	R	3 _D	3 _D
RF_ID	7:0	R	3 _D ... 1ch Tx, 3ch Rx	3 _D

The Digital_ID as well as the RF_ID will be incremented according to the latest chip release/version.

3.3 **STAT1 - Status Register1**

The status register provides internal counter values for the actual number of frames and shapes. They are also provided to the data header. However it should be mentioned that for all status registers, STAT0, STAT1, and FSTAT, with the exception of the FIFO status and error flags, updates to each status register field can happen on different timing events relative to FSM states and the field content should be treated independently from one-another. In CW mode the status bits can be read properly after eg. 100 µs.

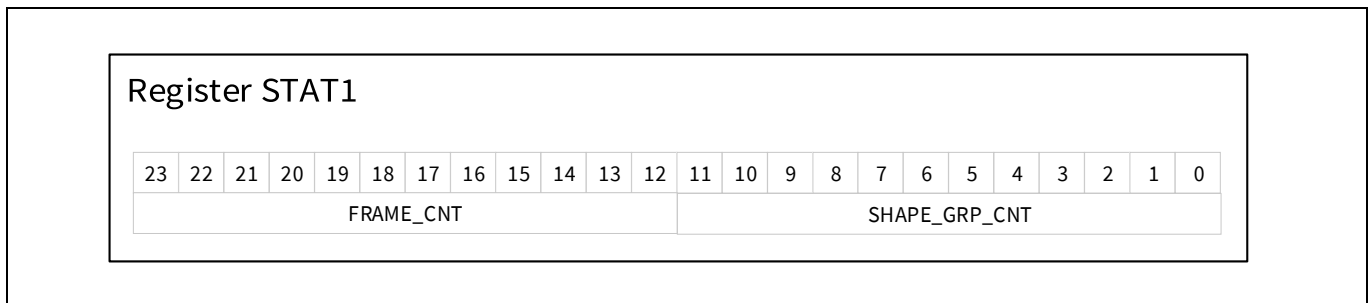


Figure 6 **STAT1: status register 1**

Table 16 **STAT1: Register Description**

Symbol	Bits	Type	Description	RST
FRAME_CNT	23:12	R	Frame counter value: 0 _D ... Reset value / after max. value rollover 4095 _D ... Max. value Note: This field is for debug only.	0 _D
SHAPE_GRP_CNT	11:0	R	Shape group counter counts the actual shape groups: 0 _D ... Reset value / after max. value for SHAPE_GRP_CNT reached 4095 _D ... Max. value	0 _D

Note:

1. A shape consists of an “Up Chirp” segment and a “Down Chirp” segment.
2. A sawtooth shape is generated by an “Up Chirp” and a “Fast Down Chirp”.
3. There is no data acquisition in the “Fast Down Chirp”.

BGT60TR13C Registers

3.4 SFCTL – SPI and FIFO Control Register

This register is used to configure the SPI and FIFO.

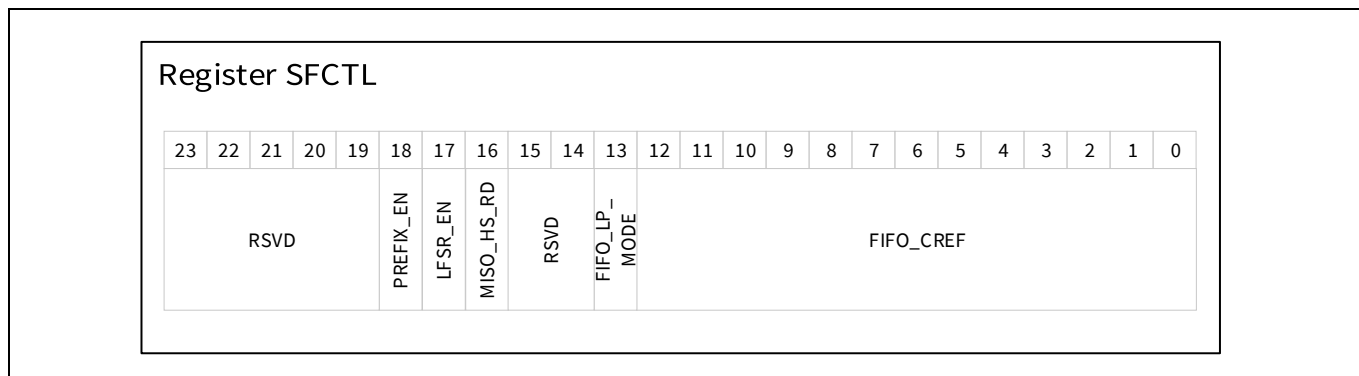


Figure 7 SPI and FIFO Control Register

Table 17 SPI and FIFO Control: Register Description

Symbol	Bits	Type	Description	RST
RSVD	23:19	RW	RSVD	14 _B
PREFIX_EN	18	RW	Enables the data header written into the FIFO prior to the sampling data of each chirp: 0 _B ... No prefix data header prior to chirp data 1 _B ... Prefix data header added prior to chirp data (see section 4.1 for data header)	0 _B
LFSR_EN	17	RW	Enable LFSR register data generation: 0 _B ... Normal data acquisition, LFSR reset 1 _B ... LFSR data generation started LFSR should be enabled after a FIFO reset to ensure an empty FIFO (see also 4.10).	0 _B
MISO_HS_RD	16	RW	0 _B ... MISO data is sent with falling edge of SPI CLK 1 _B ... MISO data is sent with rising edge (½ cycle earlier) Note: HS_RD = 0 _B can only be used for a SPI clock < 25 MHz. For HS-transfer please check the timing of the SPI Master and adjust settings accordingly. The setting becomes active when the last bit of FSCTL is clocked out and it affects MISO immediately. See also section 4.3.1.	1 _B
RSVD	15:14	RW	RSVD	0 _B
FIFO_LP_MODE	13	RW	FIFO power mode: 0 _B ... FIFO permanently enabled 1 _B ... FIFO activated dynamically	0 _B
FIFO_CREF	12:0	RW	FIFO compare reference: it defines the compare filling status for interrupt and CREF reporting	0 _B

BGT60TR13C Registers

Symbol	Bits	Type	Description	RST
			When filling status is > FIFO_CREF an interrupt is issued: $0_D \dots$ minimum value is 0, interrupt generated in case first sample is written into FIFO $8191_D \dots$ maximum value in case FIFO is full with 8192 memory locations eg. CREF = $0x1000$ represents a 50% compare reference	

3.5 STAT0 - Status Register 0

The status register STAT0 provides the actual value of some specific internal states. However it should be mentioned that for all status registers, STAT0, STAT1, and FSTAT, with the exception of the FIFO status and error flags, updates to each status register field can happen on different timing events relative to FSM states and the field content should be treated independently from one-another. In CW mode the status bits can be read properly after eg. 100 μ s.

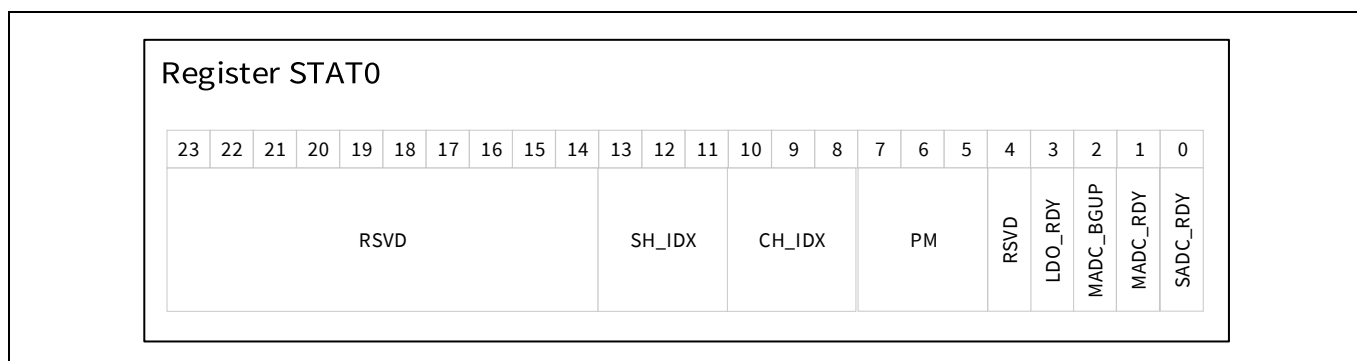


Figure 8 STAT0: Control FSM Status Register

Table 18 STAT0: Register Description

Symbol	Bits	Type	Description	RST
RSVD	23:14	R	RSVD	0_D
SH_IDX	13:11	R	Actual chirp shape enabled by the FSM: $0_D \dots$ PLLU1 $1_D \dots$ PLLD1 $2_D \dots$ PLLU2 $3_D \dots$ PLLD2 $4_D \dots$ PLLU3 $5_D \dots$ PLLD3 $6_D \dots$ PLLU4 $7_D \dots$ PLLD4	0_D
CH_IDX	10:8	R	Actual channel set enabled by the FSM: $0_D \dots$ CSU1 $1_D \dots$ CSD1 $2_D \dots$ CSU2 $3_D \dots$ CSD2 $4_D \dots$ CSU3 $5_D \dots$ CSD3	0_D

BGT60TR13C Registers

Table 18 STAT0: Register Description

Symbol	Bits	Type	Description	RST
			6 _D ... CSU4 7 _D ... CSD4	
PM	5:7	R	Power Mode is the current power mode status of FSM: 1 _D ... Active Mode 2 _D ... Interchirp Mode 3 _D ... Idle Mode 5 _D ... Deep Sleep Mode 0 _D , 4 _D , 6 _D , 7 _D ... RSVD	5 _D
RSVD	4	R	RSVD	0 _B
LDO_RDY	3	R	LDO output level, i.e. VDDC, above the threshold: 0 _B ... LDO output level below threshold 1 _B ... LDO output level above threshold, ready	0 _B
MADC_BGUP	2	R	MADC bandgap reference power up status: 0 _B ... Status down 1 _B ... Up and running	0 _B
MADC_RDY	1	R	MADC status: 0 _B ... Status down 1 _B ... Up and running	0 _B
SADC_RDY	0	R	SADC startup / calibration status: 0 _B ... Status down 1 _B ... Up and running	0 _B

3.6 FSTAT - FIFO Status Register

The global status register FSTAT is used to monitor the FIFO. It should be mentioned that for all status registers, STAT0, STAT1, and FSTAT, with the exception of the FIFO status and error flags, updates to each status register field can happen on different timing events relative to FSM states and the field content should be treated independently from one-another. In CW mode the status bits can be read properly after eg. 100 µs.

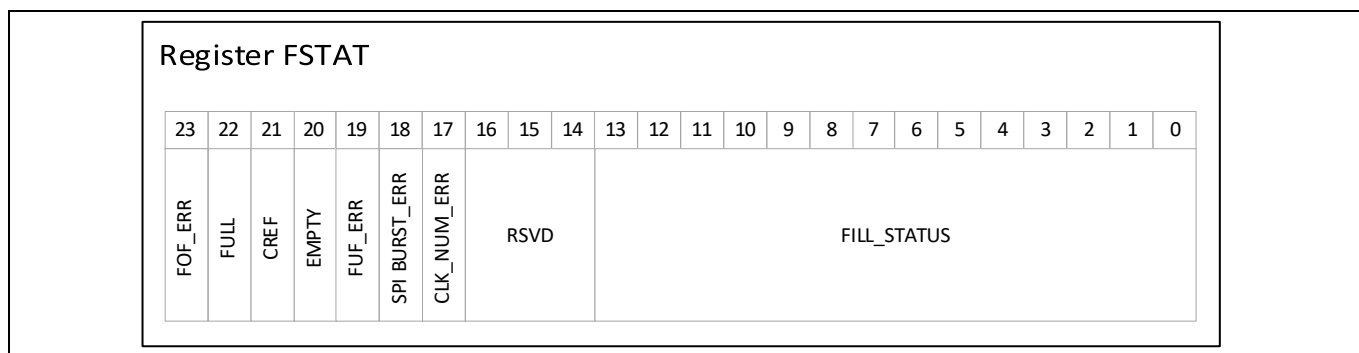


Figure 9 FSTAT Register

Table 19 FSTAT: Register Description

Symbol	Bits	Type	Description	RST
FOF_ERR	23	R	FIFO overflow error bit shows if more sample data are transferred to the FIFO than FIFO memory locations are	0 _B

BGT60TR13C Registers

Symbol	Bits	Type	Description	RST
			available to store the data. The flag will be shown also in GSR0 as a part of FIFO over or underflow error bit FOU_ERR: 0 _B ... No FIFO overflow 1 _B ... FIFO had an overflow condition	
FULL	22	R	The FULL bit shows if the FIFO has fully filled up: 0 _B ... FIFO is not full 1 _B ... FIFO is full	0 _B
CREF	21	R	0 _B ... FIFO filling status below CREF 1 _B ... FIFO filling status is > CREF	0 _B
EMPTY	20	R	The FIFO empty bit EMPTY signals if the FIFO is empty: 0 _B ... FIFO stores at least one sample 1 _B ... FIFO is empty	1 _B
FUF_ERR	19	R	FIFO under flow error signals if the host was reading more sampling data from the FIFO than available. The flag will be shown also in GSR0 as a part of FIFO over or underflow error bit FOU_ERR: 0 _B ... No error 1 _B ... FIFO underflow occurred	0 _B
SPI BURST_ERR	18	R	In case of burst error this bit is set. Further details in the Note below and in 4.8: 0 _B ... No error 1 _B ... Burst error occurred. Bit will be reset after HW or SW reset condition. See also section 4.8.	0 _B
CLK_NUM_ERR	17	R	Clock number error bit is set when SPI clocks do not fit the expected clock cycles. Further details in the Note below and in 4.8: 0 _B ... No error 1 _B ... Burst error occurred. Bit will be reset after HW or SW reset condition. See also section 4.8.	0 _B
RSVD	16:14	R	Not used	0 _D
FILL_STATUS	13:0	R	FIFO filling status: 0x0 ... FIFO empty 0x1000 ... FIFO 50% filled 0x2000 ... FIFO full This bit field is for de-bugging only. It should not be evaluated while the MADC sampling and filing up the FIFO. It can be evaluated when the FSM status is held, for example, after an FSM reset or in a specific power mode, Deep Sleep e.g.	0 _D

Note:

FOF/FUF will be cleared after these resets:

BGT60TR13C Registers

- FIFO reset
- SW reset
- HW reset

3.7 GSR0 - Global Status Register

The global status register GSR0 is related to SPI read/write monitoring.

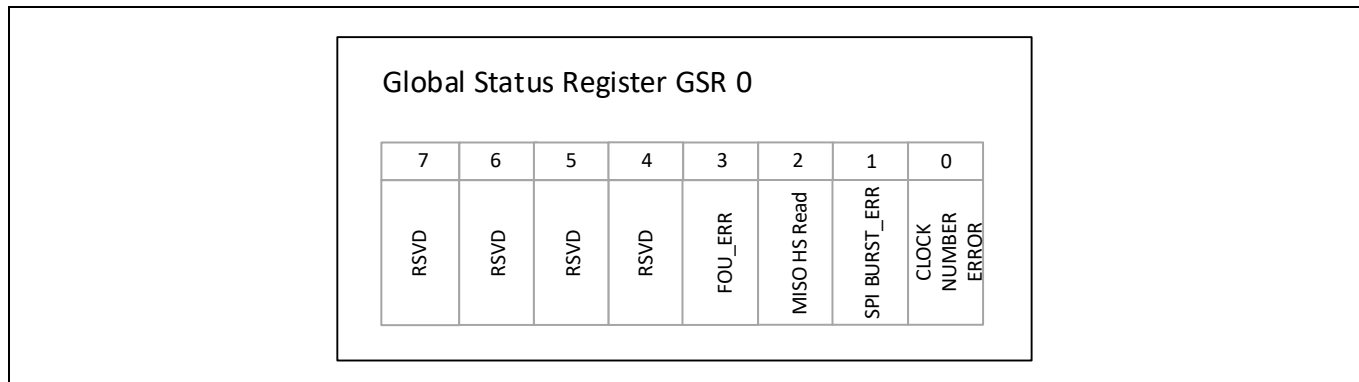


Figure 10 GSR0 Register

Table 20 GSR0: Register Description

Symbol	Bits	Type	Description	RST
RSVD	7	R	Reserved	
RSVD	6	R	Reserved	
RSVD	5	R	Reserved	
RSVD	4	R	Reserved	
FOU_ERR	3	R	Shows if FIFO overflow or underflow condition occurred. The error will be cleared after the following resets: FIFO reset or SW reset or HW reset: 1 _B ... Error condition occurred 0 _B ... No error	0 _B
MISO HS Read	2	R	SPI: MISO high speed mode activated	1 _B
SPI BURST_ERR	1	R	SPI burst error, defined within the SPI spec (see section 4.8): 0 _B ... No burst read/write error 1 _B ... Burst error	0 _B
CLOCK NUMBER ERROR	0	R	Defined within the SPI spec: 0 _B ... No clock number error 1 _B ... Error condition occurred	0 _B

Data Organization and SPI Interface

4 Data Organization and SPI Interface

4.1 Data Header

The main FSM is capable of generating a data header to be attached to the actual radar raw data. The structure of the header is shown in Figure 11 and in Table 21. The data header can be disabled by controlling the bit SFCTL:PREFIX_EN (see section 3.4).

A sync-word is sent at the beginning of each acquisition to make the radar raw-data from each shape unique. This can be useful in case of broken communication with the application processor or in case of errors. Supposing the FIFO will generate a “FIFO overflow flag” the sync-word 0x000000 can be evaluated by the host controller and used to resync with the BGT60TR13C and discard the data received before this sync word (if header or sync-word not used then the controller should reset the FIFO, discarding the actual FIFO data). On “FIFO underflow flag”, the received data bits from the host are 1111111111_B.

Following, the header includes also the frame counter and shape group counter, as well as the actual APU/APD value and temperature value.

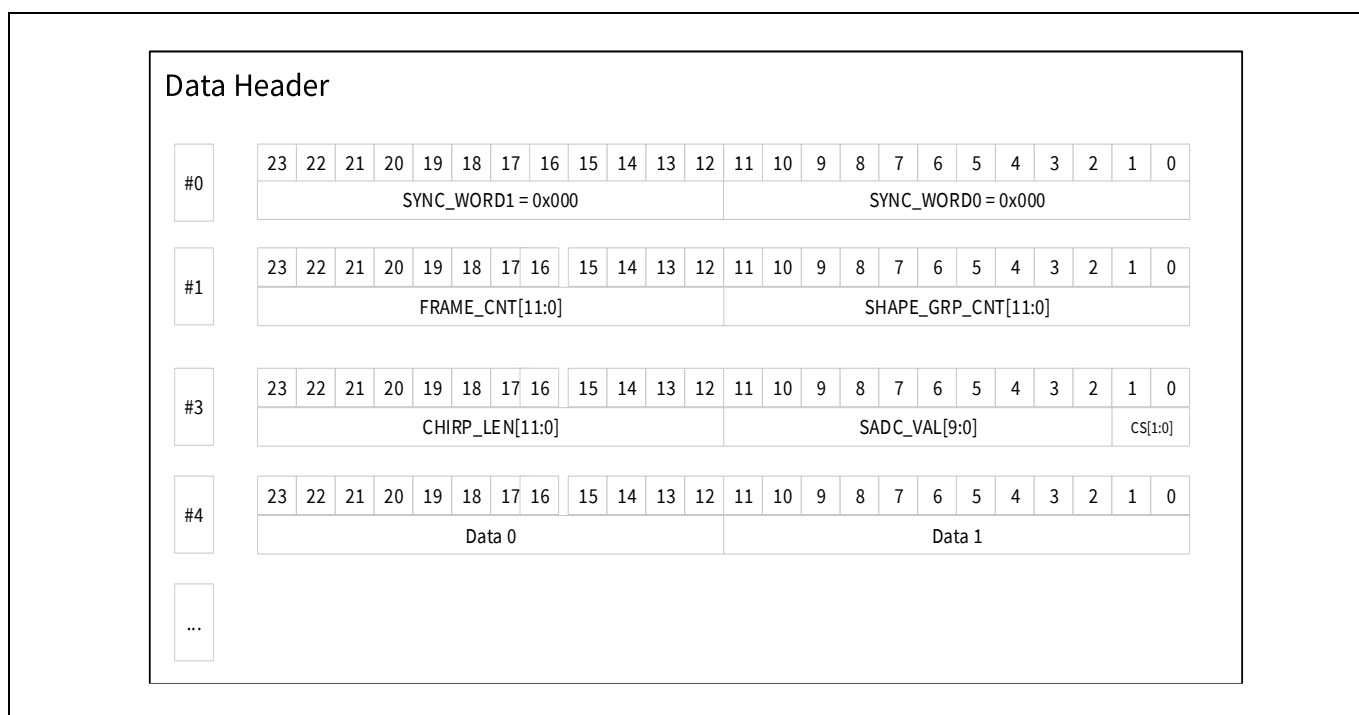


Figure 11 Data header

Data Organization and SPI Interface

Table 21 Data Header Description

Symbol	Reg	Bits	Description	RST
SYNC_WORD1	#0	23:12	The sync-word can be used to identify the start of a new chirp. In case the MADC will output also a sequence of 0x000000, to make the sync-word unique, the data from the MADC will be automatically changed to 0x001 before transferring it to the FIFO.	0 _D
SYNC_WORD0	#0	11:0	See SYNC_WORD1.	0 _D
FRAME_CNT	#1	23:12	Same as STAT1:FRAME_CNT (see section 3.3).	0 _D
SHAPE_GRP_CNT	#1	11:0	Same as STAT1:SHAPE_GRP_CNT (see section 3.3).	0 _D
CHIRP_LEN	#2	23:12	Number of MADC samples inside the chirp (APU/APD value of related chirp).	0 _D
SADC_VAL	#2	11:2	10 bits of sensor ADC output data (eg. temperature).	0 _D
CS	#2	1:0	Indicates the channel shape number to which the following sampling data belongs to.	
Data 0	#3	23:12	MSB data (see section 4.2).	
Data 1	#3	11:0	LSB data (see section 4.2).	

4.2 FIFO and Dataflow

The memory in the BGT60TR13C is based on a FIFO. The FIFO consists of a circular shift register organized in 8192 words of 24 bits each. Three dataflow modes from MADC to the FIFO are supported by the FSM (see Figure 12):

- Mode 1: Only one ADC active (can be any ADC from 1 to 3)
 - Data from 1st sample, 12 bits, are temporarily stored in a buffer
 - When the 2nd sample, 12 bits, are available, both, 1st and 2nd, 24 bits, are stored into one data word
- Mode 2: Two ADCs active (can be any ADC from 1 to 3)
 - Data from active ADCs, 12+12 bits, are occupying one data word in the FIFO
- Mode 3: Three ADCs active
 - Data from first two channels are stored into a data word while data from third channel is buffered. On the consecutive trigger the buffered data and the one from first channel are stored in a data word while the data of the second channel is buffered. On the next trigger the buffered data plus the data from the third channel are stored in a third data word, etc.

Data Organization and SPI Interface

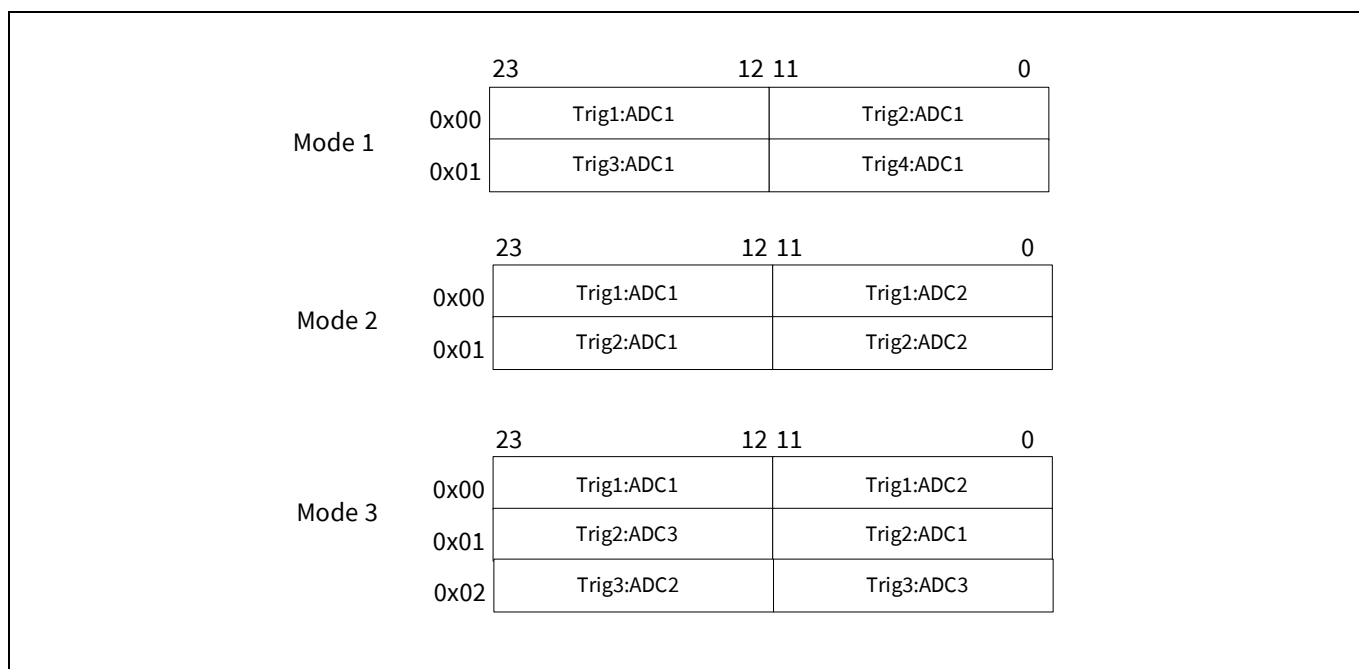


Figure 12 FIFO organization

Readout from the FIFO is done from the SPI block. Due to max frequency of SPI clock, 50 MHz, the readout rate from the FIFO is:

- $50 \text{ MHz} / \text{SPI Mode} / 24 \text{ bit} = 480 \text{ ns} = 40 \text{ cycles (in the 80 MHz domain)}$

Readout from the FIFO should be executed from the host controller using memory address with correct data length. Data length can be derived from the data header or based on the “sync-word”.

Note:

An illegal write to memory address space will lead to lost FIFO data!

4.3 SPI – Serial Peripheral Interface Module

The SPI is the communication interface between the host and the BGT60TR13C. It enables the host to read from, or write to (program) the registers as well as reading from the FIFO.

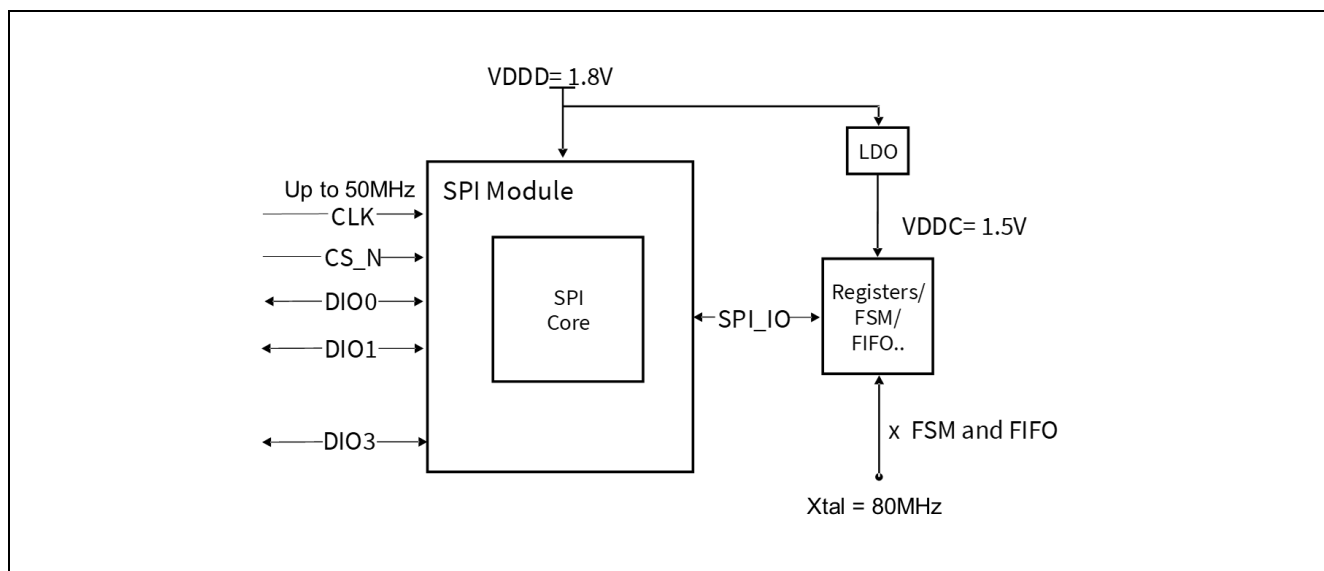


Figure 13 SPI module

Data Organization and SPI Interface

BGT60TR13C device features four I/O pins for SPI communication and one for chip reset. DIO[x] pins are pulled up to logic high inside the pad.

- CS_N to be connected to SS of the SPI master
- CLK to be connected to CLK of the SPI master
- DIO0, DI to be connected to MOSI of the SPI master
- DIO1, DO to be connected to MISO of the SPI master
- DIO2 not available on BGT60TR13C
- DIO3 to be connected to reset

Table 22 SPI Pins

Pin Name	Standard SPI Mode Function	Remarks
CS_N	CS_N	Chip select
CLK	CLK	SPI clock
DIO0	DI	HiZ, bidirectional
DIO1	DO	HiZ, bidirectional
DIO2	N.A.	Not available on BGT60TR13C
DIO3	RESET	HiZ, bidirectional

The SPI interface can be clocked up to 50MHz. To meet the timing requirements for higher SPI clock frequencies (e.g. > 25MHz) the BGT60TR13C device offers an additional high speed mode (SFCTL:MISO_HS_RD) which increase the timing budget on SPI master side by sending out data via DO with the rising edge instead of the falling edge of the CLK.

4.3.1 Standard SPI Timing

The timing diagram for normal SPI mode (SFCTL:MISO_HS_RD =0) is presented in Figure 14. A SPI transfer is started with a falling edge of chip select signal CS_N generated by the SPI master. At the same time the SPI master shall drive the level of the data input signal DI (master output, slave input) according to the first bit. Also with the falling edge of the chip select signal CS_N the SPI slave applies the level of the data output signal DO (master input, slave output) according to the first bit which shall be transferred to the SPI master, the level becomes stable after the period t(dS). The SPI master has to wait for the time t(L) before the clock signal CLK can be generated.

With the rising edge of CLK the SPI slave captures the level of DI. The SPI master must keep the DI level stable for t(mos) before and for t(moh) after the rising edge of CLK to ensure valid setup and hold time of the SPI slave. With the falling edge of CLK the SPI master shall set the level of DI according to the next bit the master wants to send.

The SPI master is supposed to read the level of DO with the rising edge of CLK. The SPI slave keeps the DO level stable for t(mis) before and for t(mih) after the rising edge of CLK. With the falling edge of CLK the SPI slave drives the level of DO according to the next bit, DO becomes stable after t(mih).

After the last bit has been transferred and CLK has gone to low level, the SPI master must set CS_N to high level to stop the transfer. The master must take care that the period between the last rising edge of CLK and the rising edge of CS_N is not shorter than t(T). Within the period t(dh) after the rising edge of CS_N the SPI slave drives DO to high impedance state again.

Data Organization and SPI Interface

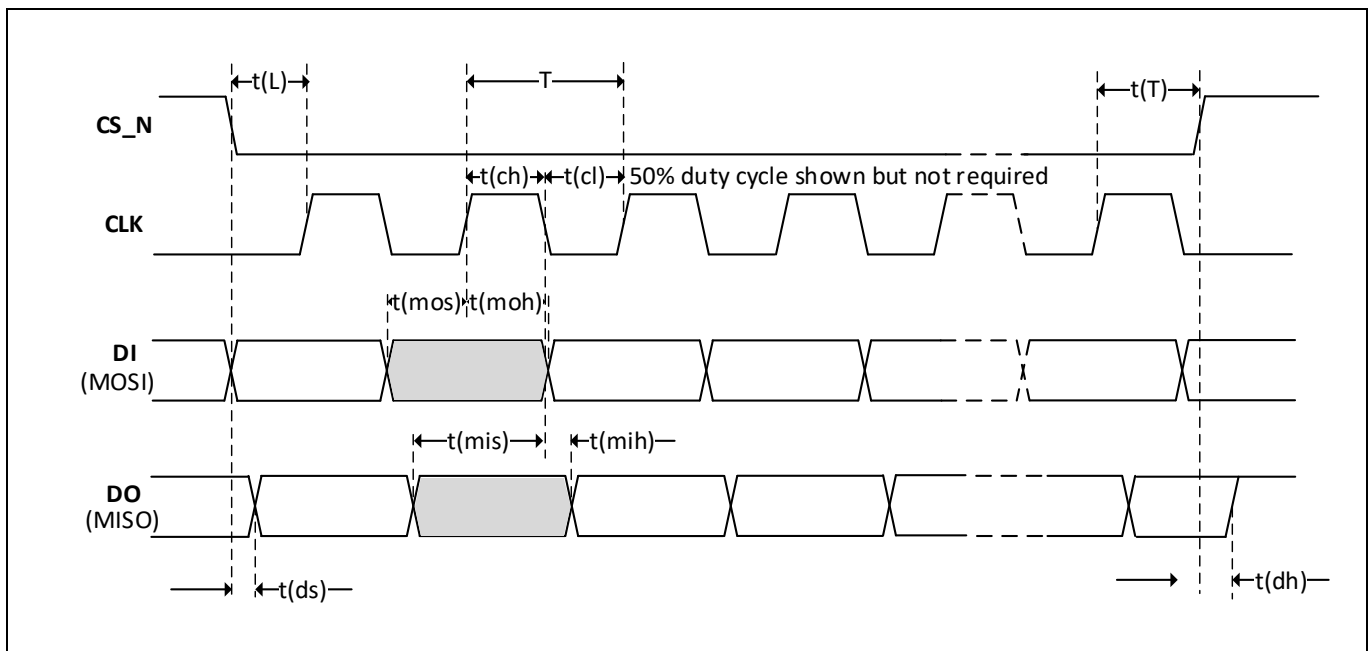


Figure 14 SPI interface timing diagram for SFCTL:MISO_HS_RD = 0_b

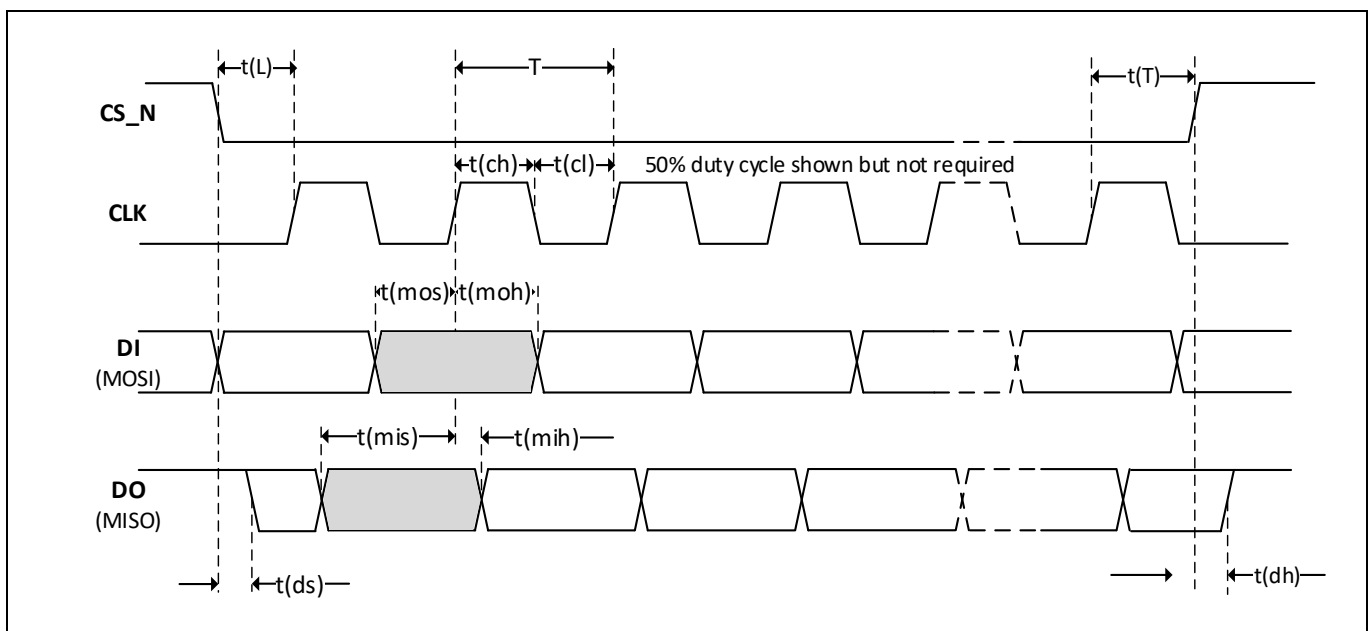


Figure 15 SPI interface timing diagram for SFCTL:MISO_HS_RD = 1_b

BGT60TR13C can operate at SPI clock frequencies up to 50MHz, but the maximum achievable SPI clock frequency is limited by DI related setup and hold times of SPI master and SPI slave. If for example the SPI master requires a longer MISO setup time than $t(mis)$, the SPI clock speed in normal SPI mode must be reduced. Alternatively BGT60TR13C can be switched to SPI high speed mode by setting SFCTL:MISO_HS_RD = 1_b.

The timing diagram for high speed SPI mode is presented in Figure 15. In this mode the SPI master is still supposed to capture the level of DO with the rising edge of CLK. The SPI slave keeps the level of DO stable for $t(mis)$ before the rising edge of CLK. The SPI slave keeps the level of DO stable for $t(mih)$ after the rising edge of CLK, and then sets the level of DO according to the next bit which is send out.

Data Organization and SPI Interface

Table 23 SPI Timing Requirements, VDDD= 1.71 to 1.89 V, Tb= -20 to +70°C

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Parameter						
SPI clock period: 50MHz, with 1% clock jitter	T	ns	20			
Clock high	t(ch)	ns	9.0			
Clock low	t(cl)	ns	9.0			
Master output setup	t(mos)	ns	5.0			
Master output hold	t(moh)	ns	5.0			
Master input setup	t(mis)	ns	5.0			T=20ns (see Note 2)
Master input hold	t(mih)	ns	1.0			
Lead time before the first working clock edge occurs	t(L)	ns	9.0			
Tailing time after the last working clock edge	t(T)	ns	1			
Data setup time after the DO goes in low impedance state	t(ds)	ns			5	Guaranteed by design
Data hold time before DO goes in hi impedance state.	t(dh)	ns			5	Guaranteed by design

Note:

1. If SFCTL:MISO_HS_RD is not set properly then data read on MISO may not be correct.
2. $t(mis)$ is specified for a maximum SPI clock frequency of 50MHz. This results in a maximum delay (time output valid) of $T-t(mis) = 15ns$ between falling edge of CLK (for MISO_HS_RD=0) and DO level becoming valid. For MISO_HS_RD=1 it's the rising edge of CLK. The timing is guaranteed for worst case condition: VDDD = 1.71 V, Tb=+70°C, output load of Cload = 50 pF.

4.3.2 Logic Levels

The digital inputs and outputs are fully CMOS compatible (reported in Table 24). All IO input / output timings are based on 50% voltage reference levels (see Figure 16). I/O interfaces are shown in Figure 19, input pins, and Figure 20, output pins, which include internal pull-ups.

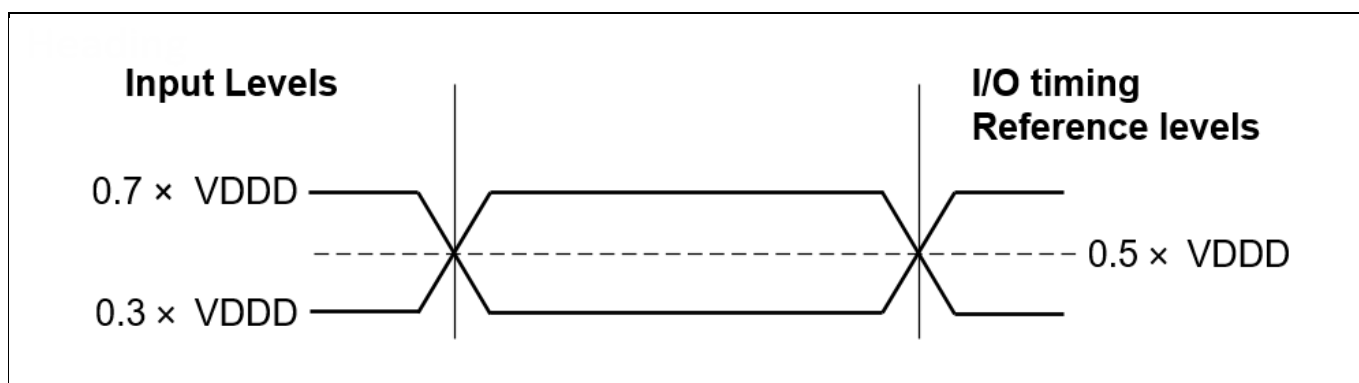


Figure 16 AC timing input/output reference levels

The input logic hysteresis prevents input buffers from oscillation. The minimum hysteresis range V_{HYST} is in between the lower ($0.3 \times V_{DDD}$) and upper logic level ($0.7 \times V_{DDD}$) boundaries (see Figure 17). Above $0.7 \times V_{DDD}$

Data Organization and SPI Interface

the input signal is a logical '1' while below $0.3 \times V_{DD}$ it is a logical '0' regardless of hysteresis. Due to temperature drifts and device variation the hysteresis range V_{HYST} can be up to $0.7 \times V_{DD}$ or down to $0.3 \times V_{DD}$ but typically around $0.5 \times V_{DD}$. Parameters reported in Table 24 and Table 25.

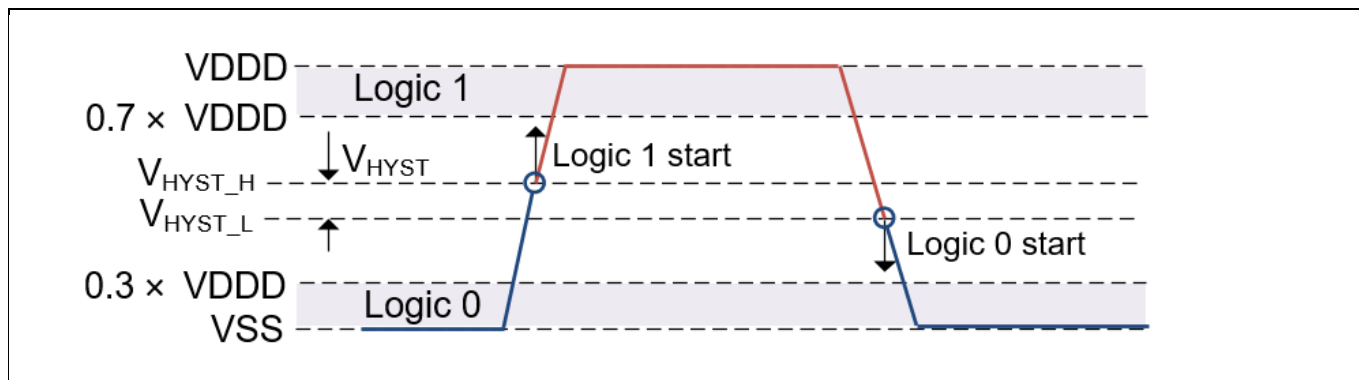


Figure 17 Logic input levels and hysteresis

The digital output pads have a fixed output pad strength that gives a specific slew-rate for rising signals, dV_{TR} , and falling signals, dV_{TF} (see Figure 18). Minimum slew rates were simulated considering a total capacitive load of 15pF. Results reported in Table 24 and Table 25.

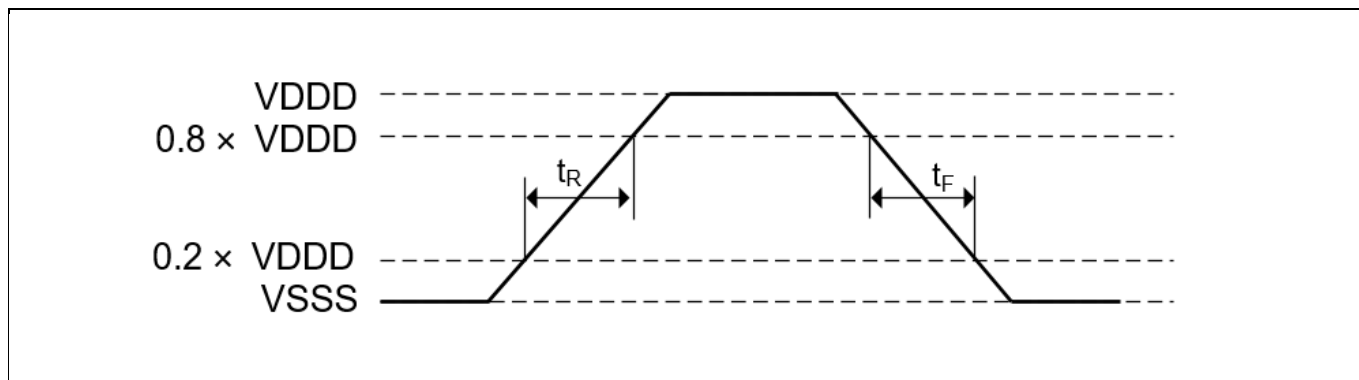


Figure 18 Rise/Fall Time, Slew Rate specified between $0.2 \times V_{DD}$ and $0.8 \times V_{DD}$

Data Organization and SPI Interface

Table 24 Logical Levels for Pins CLK, CS_N, DI, DIO3 VDDD= 1.71 to 1.89 V, Tb= -20 to +70°C, ambient temperature not below -40°C; all voltages with respect to VSSD digital ground, positive current flowing into pin (unless otherwise specified)

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Parameter			Min	Typ	Max	
LOW level	$V_{IN(L)}$	V	0		$0.3 \times V_{DDD}$	
HIGH level	$V_{IN(H)}$	V	$0.7 \times V_{DDD}$		VDDD	
Input current ($0V < V_{IN} < V_{DDD}$)	I_{IN}	μA	-150		150	
Input capacitance CLK/CS_N	C_{IN}	pF	1.8			
Input capacitance DI/DIO3	C_{IN}	pF	3.1			
Minimum hysteresis voltage range between $0.3 \times V_{DDD}$ and $0.7 \times V_{DDD}$	V_{HYST}	V	0.175			$V_{HYST_H} - V_{HYST_L}$
Upper hysteresis signal level	V_{HYST_H}	V		$0.5 \times V_{DDD} + \frac{V_{HYST}}{2}$	$0.7 \times V_{DDD}$	
Lower hysteresis signal level	V_{HYST_L}	V	$0.3 \times V_{DDD}$	$0.5 \times V_{DDD} - \frac{V_{HYST}}{2}$		
Output pad slew rate for rising wave form	dV_{TR}	V/ns	0.32			$0.2 \times V_{DDD}$ to $0.8 \times V_{DDD}$
Output pad slew rate for falling wave form	dV_{TF}	V/ns	0.33			$0.2 \times V_{DDD}$ to $0.8 \times V_{DDD}$

Data Organization and SPI Interface

Table 25 Logic Levels for Pins IRQ, DO, DIO3 VDDD= 1.71 to 1.89 V, Tb= -20 to +70°C, ambient temperature not below -40°C; all voltages with respect to VSSD digital ground, positive current flowing into pin (unless otherwise specified)

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
LOW level	$V_{OUT(L)}$	V	0		$0.3 \times V_{DDD}$	
HIGH level	$V_{OUT(H)}$	V	$0.7 \times V_{DDD}$		VDDD	
Output current (LOW)	$I_{OUT(L)}$	mA	-5			
Output current (HIGH)	$I_{OUT(H)}$	mA			5	
Allowed load capacitance to provide maximum signal frequency on DO	C_{LOAD}	pF			15	
Minimum hysteresis voltage range between $0.3 \times V_{DDD}$ and $0.7 \times V_{DDD}$	V_{HYST}	V	0.175			$V_{HYST_H} - V_{HYST_L}$
Upper hysteresis signal level	V_{HYST_H}	V		$0.5 \times V_{DDD} + \frac{V_{HYST}}{2}$	$0.7 \times V_{DDD}$	
Lower hysteresis signal level	V_{HYST_L}	V	$0.3 \times V_{DDD}$	$0.5 \times V_{DDD} - \frac{V_{HYST}}{2}$		
Output pad slew rate for rising wave form	dV_{TR}	V/ns	0.32			$0.2 \times V_{DDD}$ to $0.8 \times V_{DDD}$
Output pad slew rate for falling wave form	dV_{TF}	V/ns	0.33			$0.2 \times V_{DDD}$ to $0.8 \times V_{DDD}$

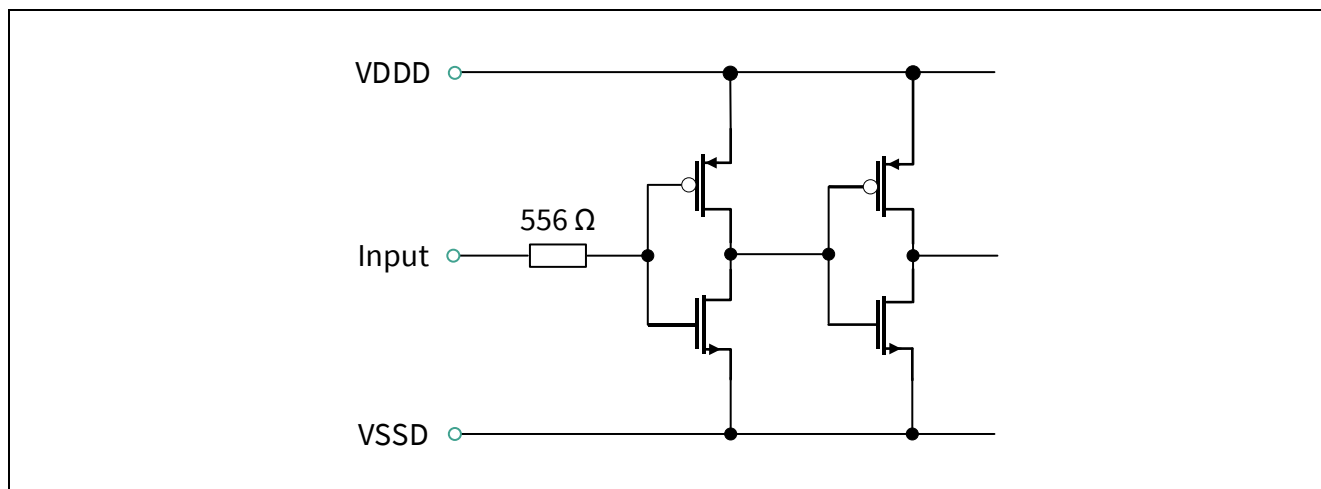


Figure 19 Interface for input pins CLK, CS_N, DI, DIO3

Data Organization and SPI Interface

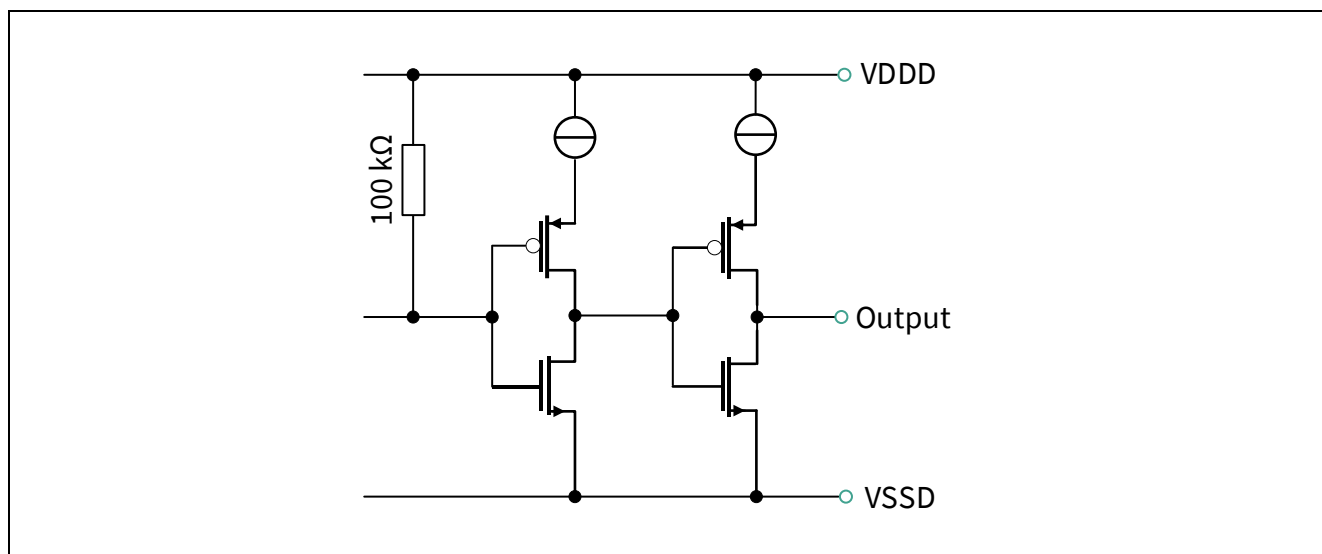


Figure 20 Interface for output pins IRQ, DO, DIO3

4.4 Overshoot and Undershoot Waveform Definition

During operation the applied signals and supply levels should not exceed absolute maximum DC levels specified in datasheet. Digital signals can have positive or negative overshoots due to inductive and/or capacitive loads. The following **Error! Reference source not found.** Table 26 reports the allowed overshoot timings and signal levels for all logic signals.

Table 26 Overshoot and Undershoot Signal Levels

Spec	Symbol	Unit	Value			Condition
			Min	Typ	Max	
Maximum absolute overshoot voltage level	Vos	V			VDDD + 0.5 V	see Note:
Maximum absolute undershoot voltage level	Vus	V			VSS - 0.5 V	see Note:3

Note:

Maximum pad current not exceeding ± 5 mA (see also Table 25). No slew rate limitation existing on digital signals for overshoots / undershoots.

4.5 IBIS Model

A BGT60TR13C IBIS Model is available under NDA upon request. It is based on timing simulations. In order to better reflect the real timing behavior, different pad models for input/output signals are used and summarized in Table 27. The driver strength for all pads are fix (PRG0=0).

Table 27 IBIS Pad Types and Models (see Ibis model)

Pin	IBIS PAD Model
CSN	IN: MODEL_654_7345_110
CLK	IN: MODEL_654_7345_110
DIO0 / DI	IN: MODEL_8138_4982_52 OUT: MODEL_8138_4982_59

Data Organization and SPI Interface

Pin	IBIS PAD Model
DIO1 / DO	IN: MODEL_8138_4982_52 OUT: MODEL_8138_4982_59
DIO2	Not available on BGT60TR13C
DIO3 / Reset	IN: MODEL_8138_4982_52 OUT: MODEL_8138_4982_59
IRQ	OUT: MODEL_8138_4982_55

4.6 SPI Functionality

Each word transferred over the SPI bus has a length of 1 command byte + 3 data Bytes. The communication is done bitwise. First the address is transferred with MSB first. The address is followed by the R/W-bit and then followed by the data which is sent MSB first, too. At the same time, while command byte is received, a freely from system level configurative global status register (8 bits, GSR0) is serial shifted out on DO (MSB first). On the following 24 clock cycles the selected register content is shifted out on DO, MSB first.

Depending on sent R/W-bit there are two different operation modes available, the write mode and the read mode. Every write mode is a read mode too.

Write-Mode

After the start condition the desired address is sent. The address is 7 bits long followed by a bit that is a data direction bit (read/write). A one indicates a write operation (see Figure 21).

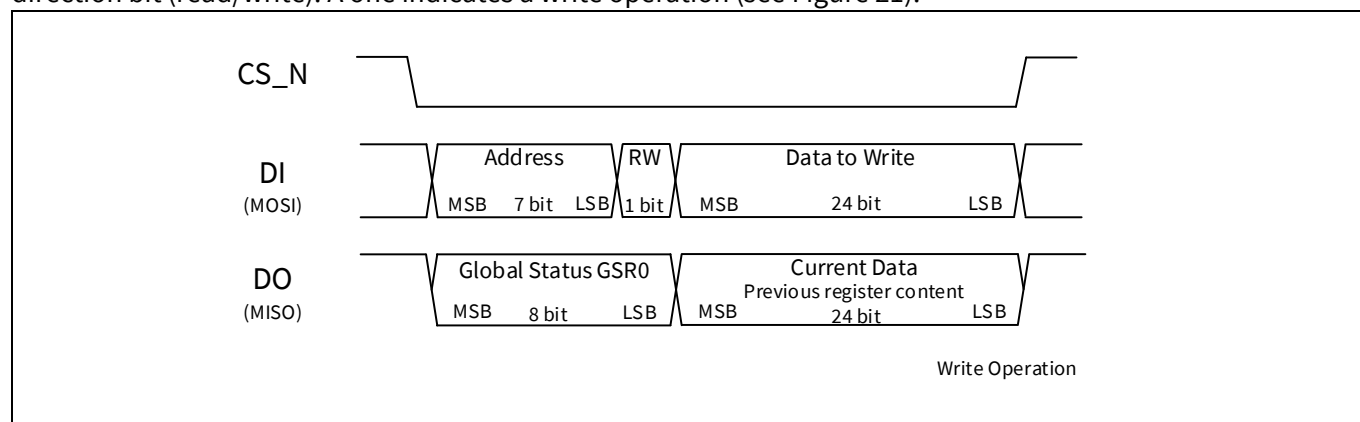


Figure 21 SPI timing write mode

Read-Mode

After the start condition, the desired address is sent like in the write operation. A zero of the R/W-bit indicates a read access. The data on DI after the command byte may contain any value. The DO behavior is the same as in write mode.

Data Organization and SPI Interface

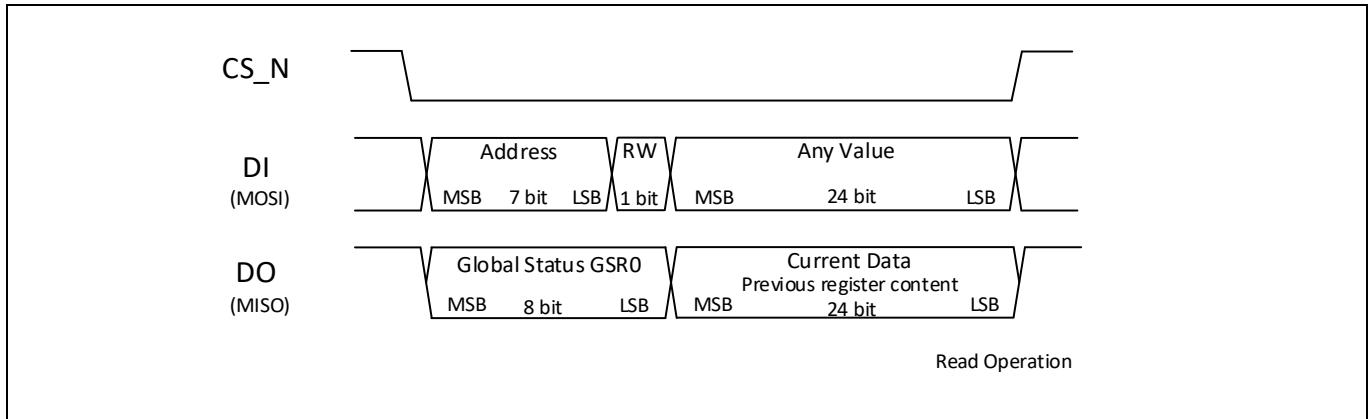


Figure 22 SPI timing read mode

4.7 SPI Burst Mode

The burst mode can be used to read or write out several registers or some data from the FIFO instead of reading just single registers or data. The burst mode command is sent by the host. The burst mode command consists of several bit fields and is shown in Figure 23.

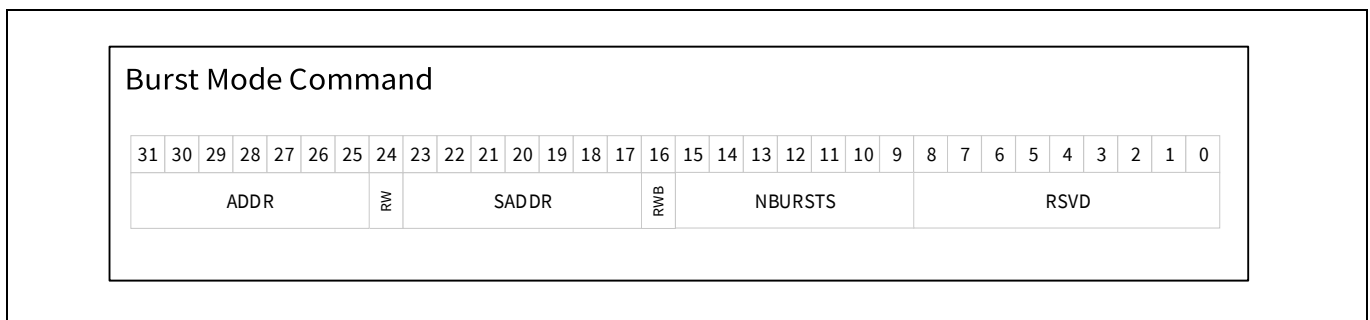


Figure 23 Burst mode command

The following Table 28 shows a detailed description on the burst mode command bit fields.

Table 28 Burst Mode Command Bit Field Description

Bit field	Bit	Description	RST
ADDR	31:25	To enter the burst mode the following address is used: 0x7F ... request the burst read/write.	
RW	24	Read/Write register access: Fixed to 1 _B ... write to address 0x7F	
SADDR	23:17	Starting address where the burst starts processing: < 0x60 Register access == 0x60 FIFO access > 0x60 Reserved - Address is incremented automatically inside a burst.	
RWB	16	Burst read or write: 0 _B ... Perform a read burst 1 _B ... Perform a write burst, (writes to FIFO not supported)	
NBURSTS	15:9	Number of processed data blocks: 0x00... “unbounded” burst accesses	

Data Organization and SPI Interface

Bit field	Bit	Description	RST
		0x01 to 0x7E ... number of words to transfer	

Note:

A single data block is 24 bits width for both, the sampling memory and the registers.

Burst Mode Operation

After the start condition the 32 bits burst mode command is sent from the SPI master on DI. At the same time, the status register GSR0 (four 1_B bits + four status bits) followed by 24 padding bits set to 0B is shifted out on DO. After the command sequence is done, the register/FIFO data is shifted out to the SPI master on DO. In burst write mode, the register data to be written is shifted in from the SPI master (application processor e.g.).

Burst Mode Read Sequence:

In the read sequence, the SPI master reads from the device.

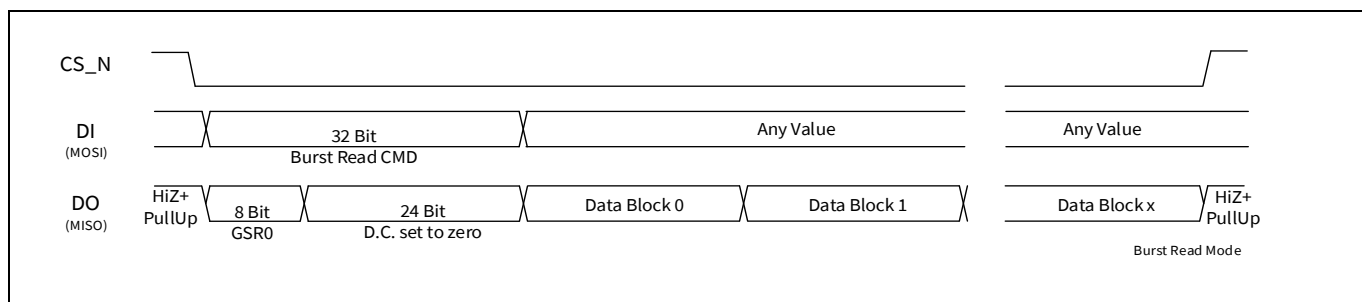


Figure 24 Burst mode read sequence

Burst Mode Write Sequence:

In the burst write mode, the SPI master writes to the device.

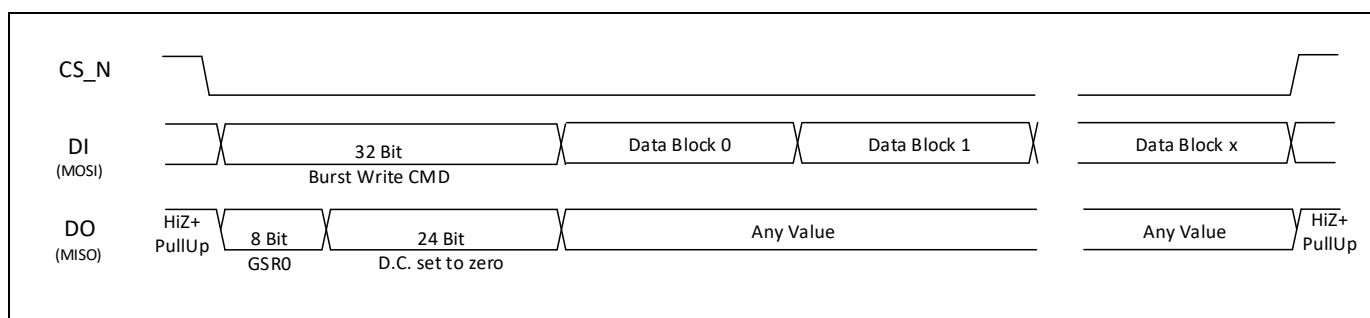


Figure 25 Burst mode write sequence

Sampling Data Arrangements in Data Blocks

The data from the FIFO are streamed out during the burst read request, starting from the FIFO address zero. The 1st ADC is the ADC channel with the lowest channel number. As far as the sampling memory is organized in 24 bits and up to three ADC channels are selectable through the ADC channel selection bits, the data blocks are arranged as follows.

In case a single ADC is selected the data blocks are shown in Figure 26.

Data Organization and SPI Interface

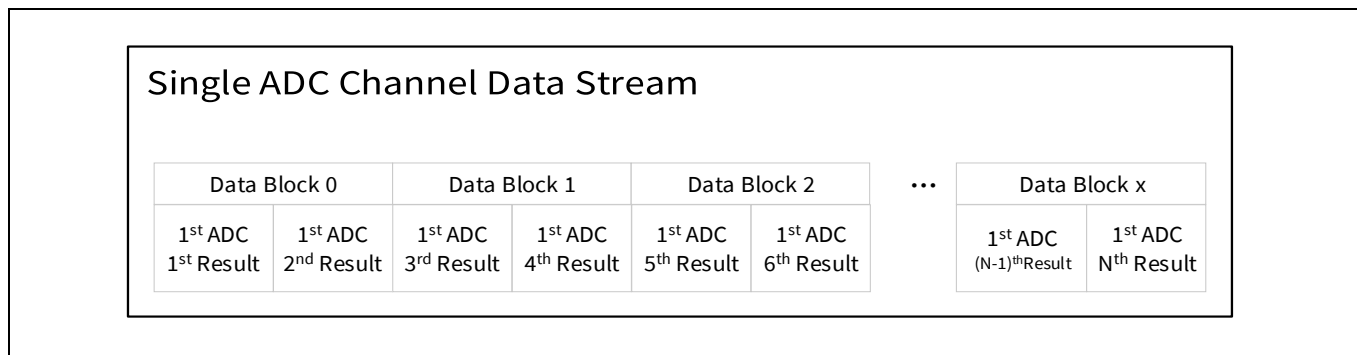


Figure 26 Single ADC channel selected

In case two ADCs are selected the data blocks are arranged as shown in Figure 27.

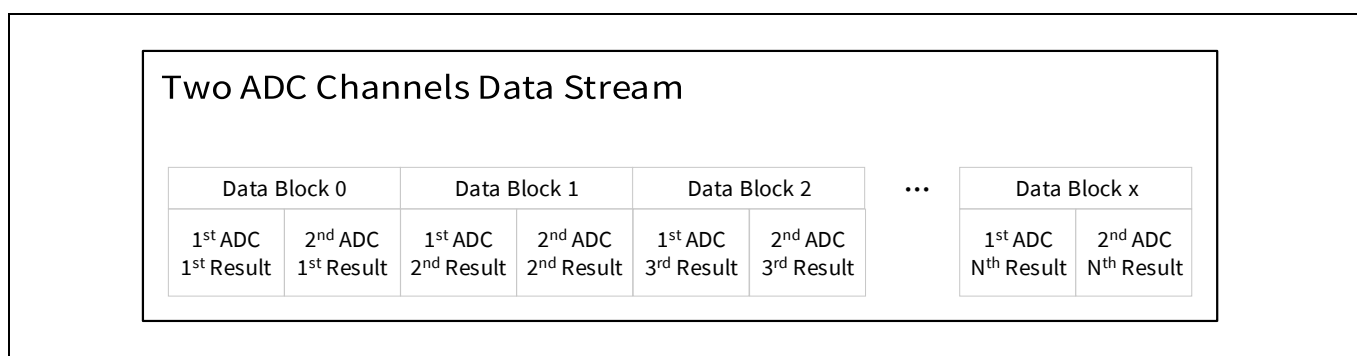


Figure 27 Two ADC channels selected

In case three ADCs are selected the data blocks are arranged as shown in Figure 28.

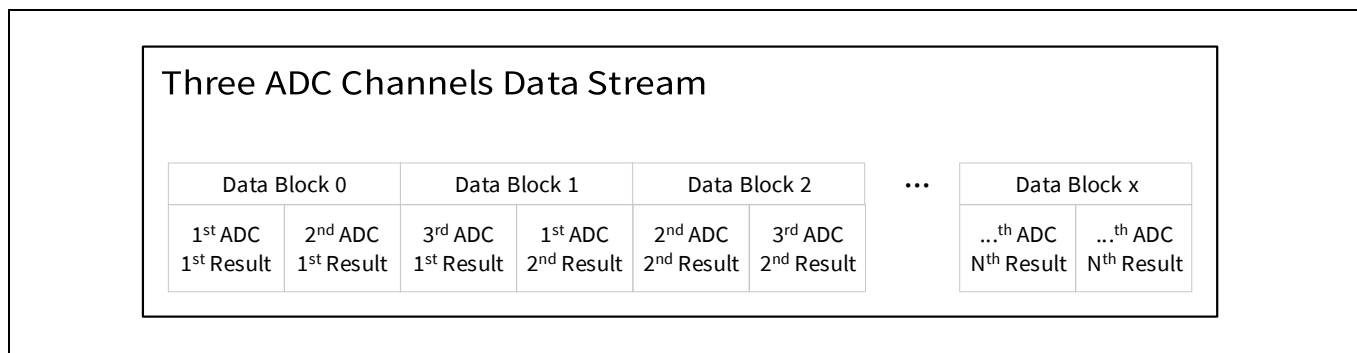


Figure 28 Three ADC channels selected

Example: Burst Mode Read Sampling Memory Sequence

The following burst mode command is sent from the host to initialize the burst mode to read from the FIFO an undefined number of sampling data:

BMCMD_RS = (ADDR = 0x7F, RW = 0x01, SADDR = 0x60, RWB = 0x0, NBURSTS = 0)

REMARK:

For each burst read request to the sampling memory, the sampling-memory address pointer is reset to the initial value. So that memory can be read out from the beginning until the application processor stops burst reading.

Data Organization and SPI Interface

Example: Burst Mode Read Registers Sequence

The following burst mode command is sent from the host to initialize the burst mode to read out 10 registers starting from register address 3:

BMCMD_RR10 = (ADDR = 0x7F, RW = 0x01, SADDR = 0x3, RWB = 0x0, NBURSTS = 10)

4.8 SPI Error Detection

SPI BURST_ERR and CLK_NUM_ERR (see also Table 19) will be cleared after these resets:

- SW reset
- HW reset

SPI BURST_ERR and CLK_NUM_ERR are reported in the global status bits of the next SPI transaction and latched as sticky bits in the FSTAT register.

In order to understand if the captured sample data are corrupted, the host can evaluate the bit field CLK_NUM_ERR and SPI BURST_ERR as reported in Table 29.

Table 29 SPI BURST_ERR and CLK_NUM_ERR Definitions

Length Range	Transaction	SPI BURST_ERR	CLK_NUM_ERR	Behavior on read/write
0	Null command	0 _B	0 _B	Ignored
1-31	Short length error in single	0 _B	1 _B	Command ignored
>32	Long length error in single	0 _B	1 _B	Extra bits ignored
1-31	Short length error in SPI burst header	0 _B	1 _B	Command ignored
<24xN	Missing whole data word in bounded burst	1 _B	0 _B	Available data words used
>24xN	Extra whole data word in bounded burst	1 _B	0	Extra data word(s) ignored
%24>0	Misaligned bit-count for bounded burst	1 _B	1 _B	Extra bits ignored
%24>0	Misaligned bit-count for infinite burst	0 _B	1 _B	Partial data word may be discarded

Note:

- *Ignored write transaction means that no register (or memory) content is affected by the partial write command, or incomplete data word.*
- *Ignored read transaction means that the returned data is invalid, and for the FIFO no words are removed by the partial read command, or incomplete data word.*
- *Discarded read transaction means that the data is already read from the FIFO but only partially transferred; subsequent read pops next word from FIFO.*
- *Data from the FIFO may be discarded after a length error in the infinite burst (NBURST=0) occurs. The FIFO read has to happen, since at that stage the data is required to be shifted out, but if not all bits are shifted out the FIFO is already read and the partial data word may be discarded.*

Data Organization and SPI Interface

4.9 Hardware Reset Sequence

The After power up, the chip is not in a default reset condition and requires a dedicated HW reset as described below. Only after the HW reset also other reset sequences can be triggered via SPI (e.g. SW, FIFO and FSM- reset). For such SPI triggered resets an external OSC_CLK (see Table 2) needs to be applied, while the HW reset does not require any external OSC_CLK.

HW Reset Sequence: While CS_N is '1_B' DIO3 must perform a 1_B → 0_B → 1_B transition

The behavior is presented in Figure 29 with:

T_CS_BRES = 100 ns

T_RES = 100 ns

T_CS_ARES = 100 ns

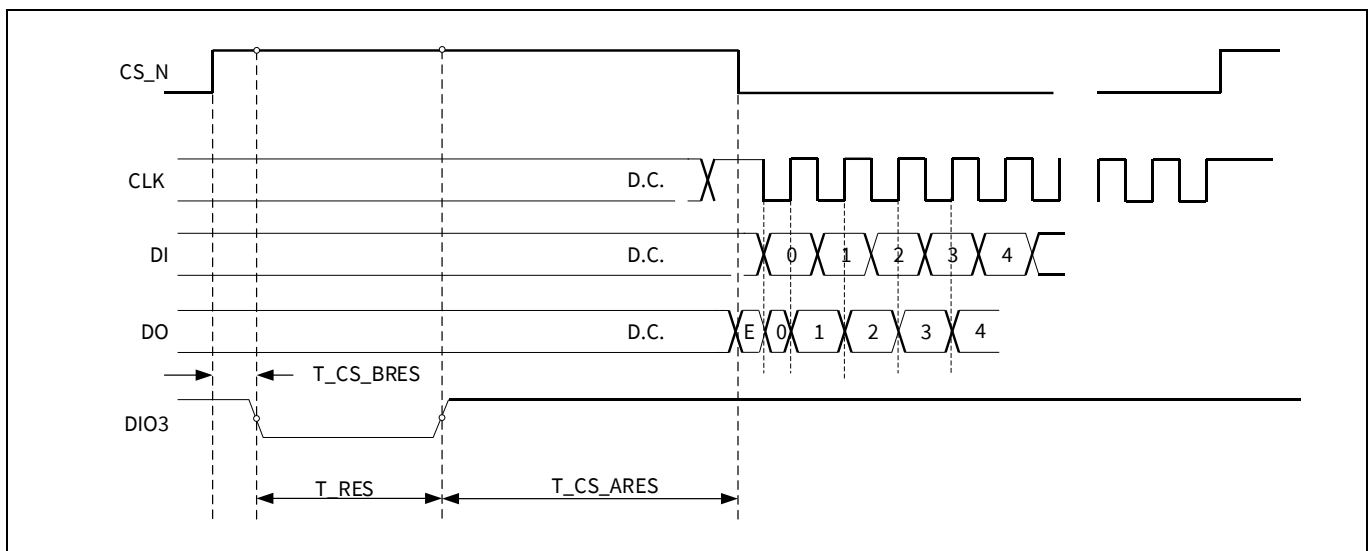


Figure 29 Hardware reset Sequence

4.10 Software Triggered Resets

Besides the hard reset, three reset sequences are supported and can be triggered in the ISO register. They are defined according to the following hierarchy:

Soft reset → FIFO reset → FSM reset

* Software Reset

- Resets all registers to default state
- Resets all internal counters (shape, frame e.g.)
- Perform FIFO reset
- Performs FSM reset
- A delay of 100 ns after the SW_RESET is needed before the next SPI command is sent

* FIFO Reset

- Reset the read and write pointers of the FIFO

Data Organization and SPI Interface

- Array content will not be reset, but cannot be read out
- FIFO empty is signaled, filling status = 0
- Resets register FSTAT
- Performs an implicit FSM reset

* FSM Reset

- Resets FSM to deep sleep mode
- Resets FSM internal counters for channel/shape set and timers
- Resets STAT0 and STAT1 register
- Reset PLL ramp start signal
- Reset PA_ON
- Terminates frame (shape and frame counters incremented although maybe not complete)

4.11 IRQ Output

BGT60TR13C provides one interrupt pin output (IRQ). In default mode, the IRQ signal is used to monitor the filling level of the FIFO as described below.

IRQ status definition:

- IRQ is high after:
 - CS_N goes high and FSTAT:FILL_STATUS $\geq$ SFCTL:FIFO_CREF (see also 3.4 and 3.6).
- IRQ is low (as a consequence of):
 - CS_N goes high and FSTAT:FILL_STATUS $<$ SFCTL:FIFO_CREF (see also 3.4 and 3.6).
 - or CS_N is active low

The following figure shows the IRQ signal in case of FIFO-burst reads.

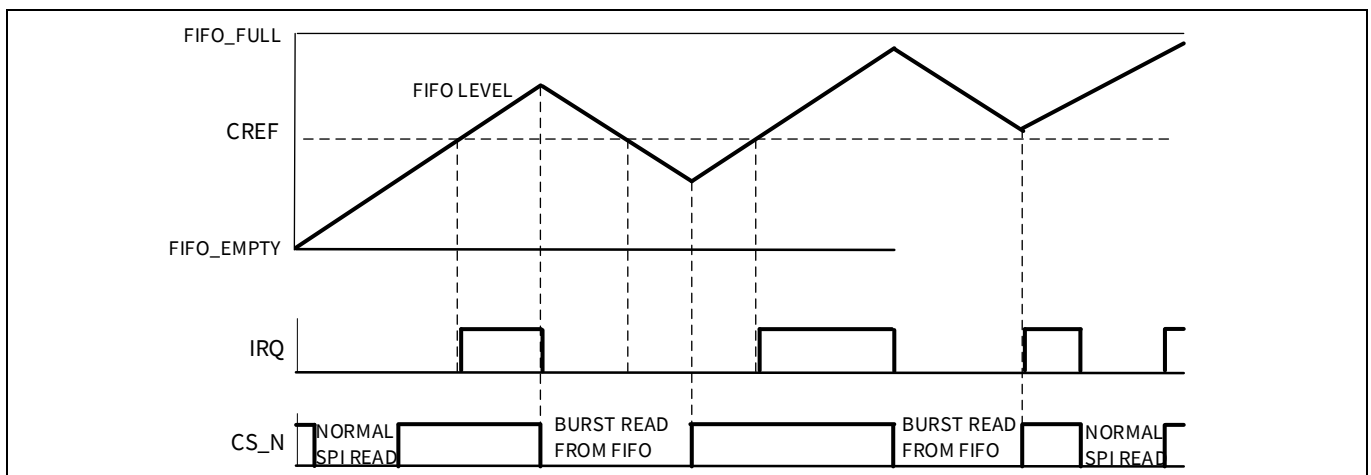


Figure 30 IRQ status behavior during radar mode with FSM capturing data

Package

5 Package

The BGT60TR13C chipset is housed in a laminate package with solder balls of 300 μm diameter and a standoff of 240 μm . According to IPC/JEDEC's J-STD0, the moisture sensitivity level (MSL) is 3. Figure 31 shows the top view of BGT60TR13C package and its physical dimension. The bottom view is presented in Figure 32. The package size is 6.5 x 5 x 0.9 mm³ with ball pitch of 500 μm . Package outline is reported on Figure 33. Package name: PG-VF2BGA40-1.

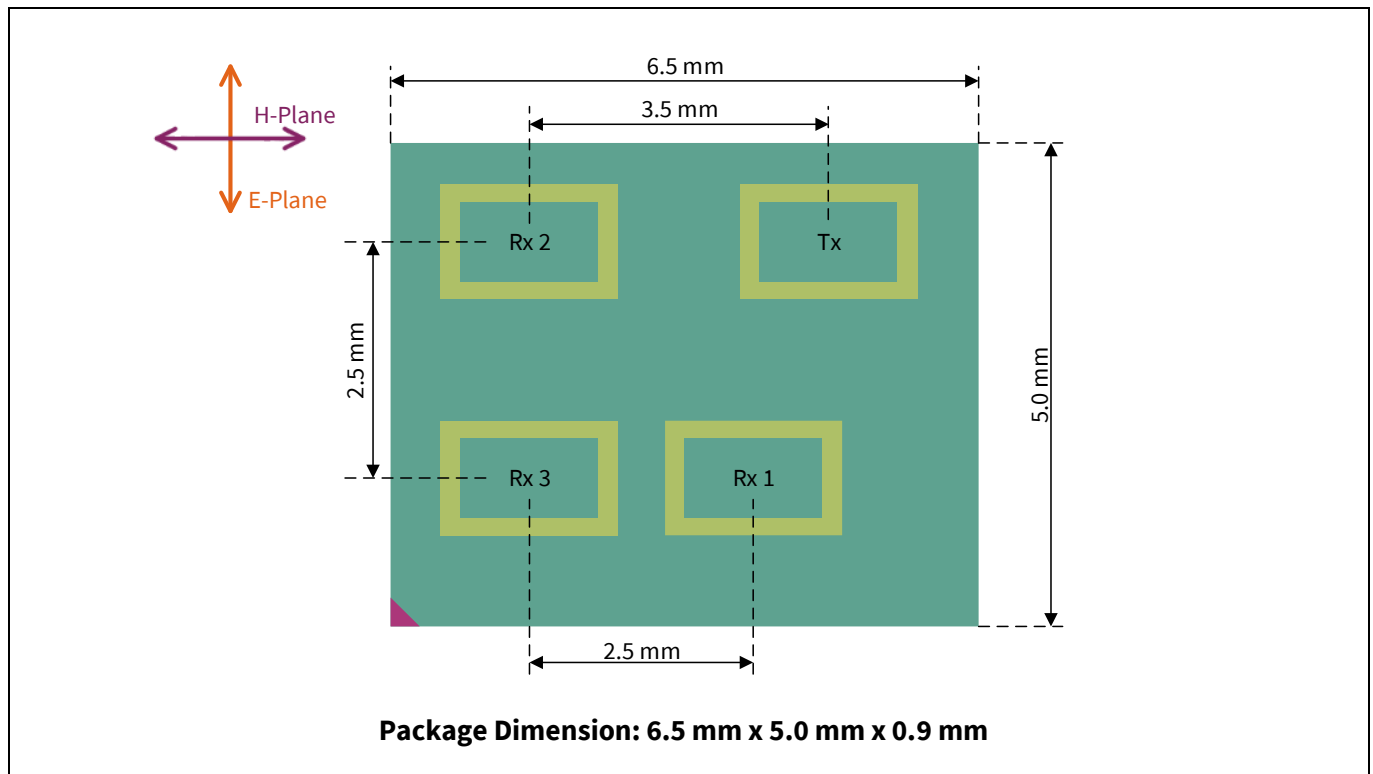


Figure 31 Top view of BGT60TR13C

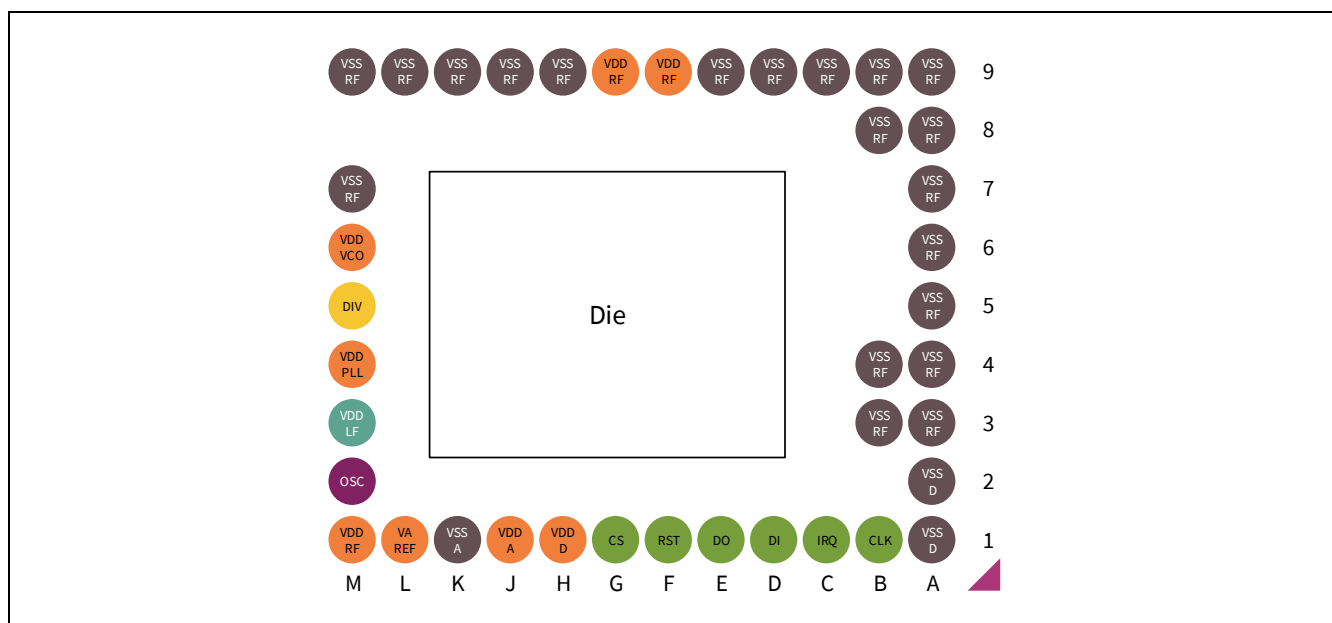


Figure 32 Bottom view of BGT60TR13C

Package

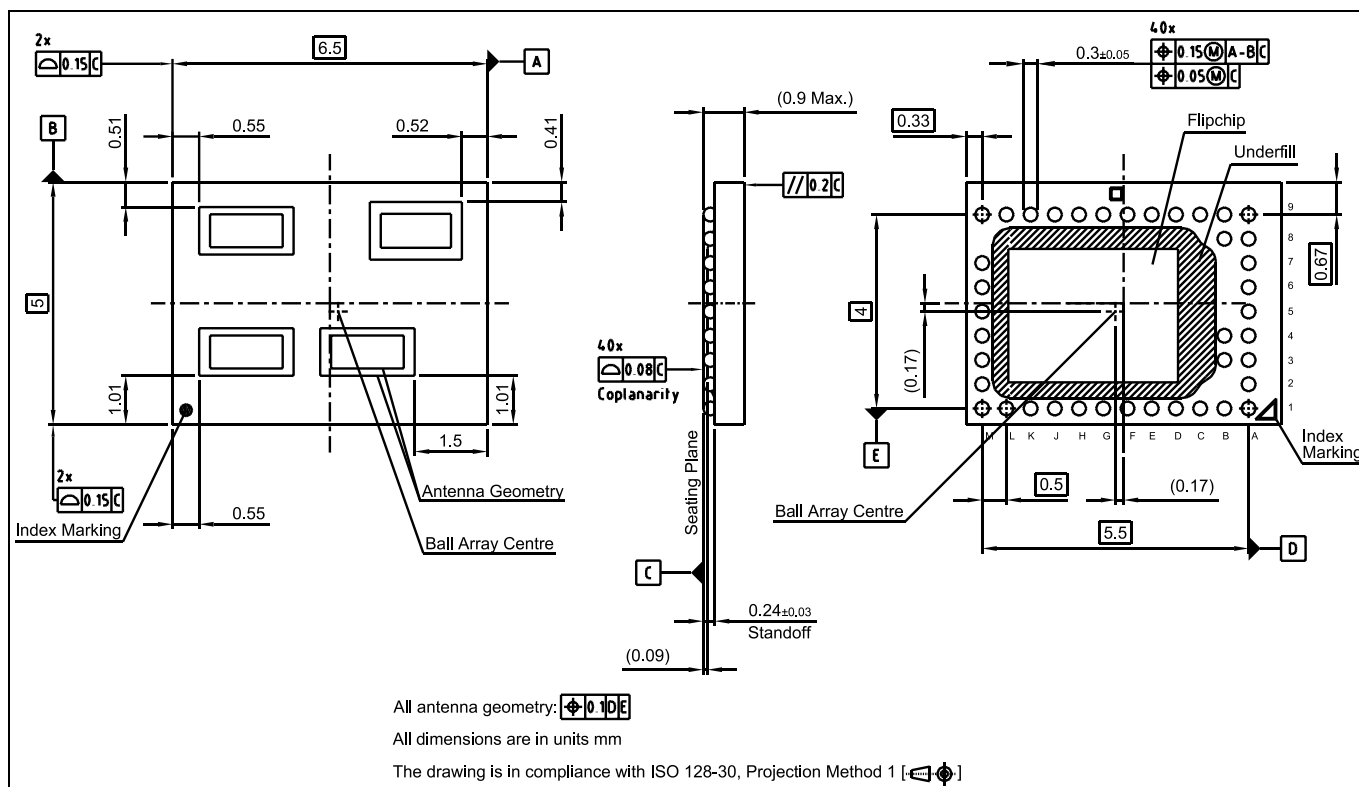


Figure 33 Package outline PG-VF2BGA-40-1

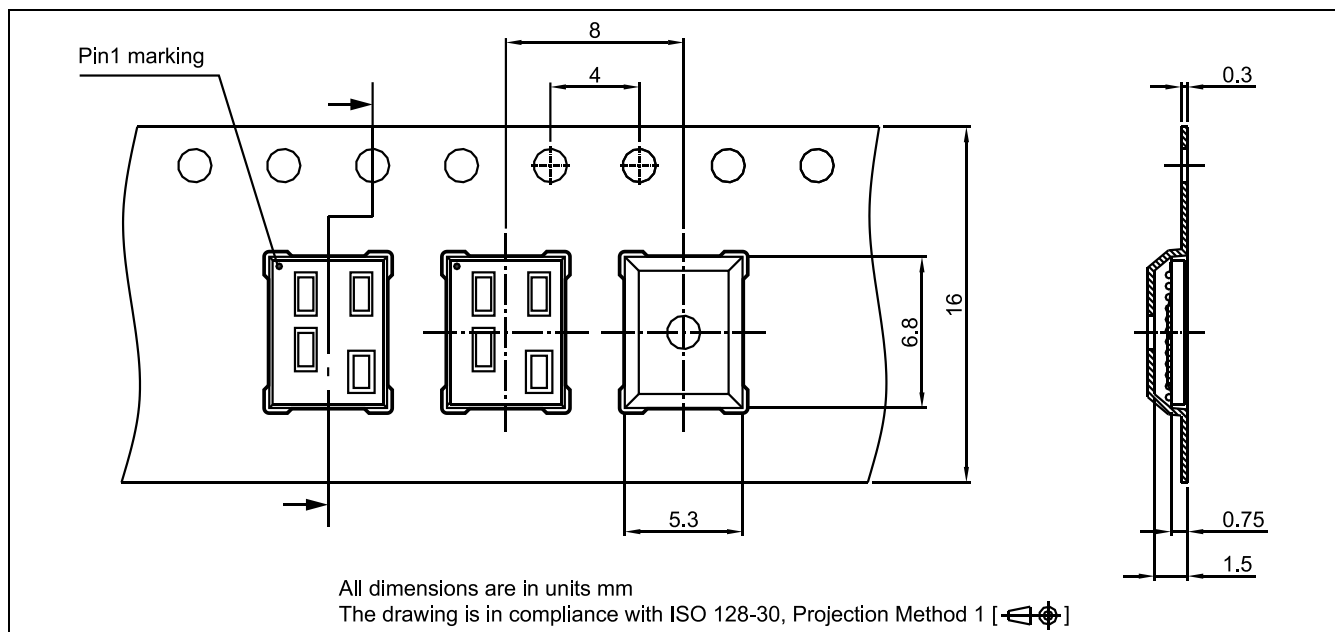


Figure 34 Tape of PG-VF2BGA-40-1

Package

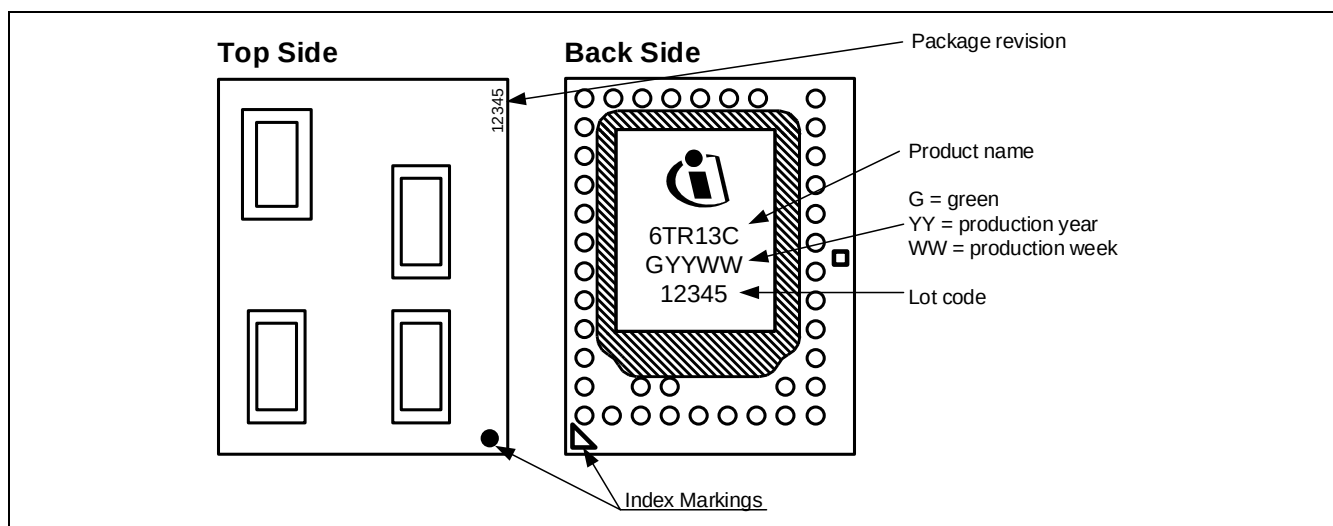


Figure 35 Marking layout of PG-VF2BGA-40-1 (example)

5.1 Built-in Antenna Specifications

Antenna performance reported in Table 30 are guaranteed by design. Typical antenna behavior is measured on Infineon reference board. Typical antenna beam plots are available in a specific application note and upon specific request.

Table 30 Antennas In Package Specifications

Spec	Unit	Value			Condition
		Min	Typ	Max	
Parameter					
RX_BW, TX_BW	GHz	58.0		63.5	Antenna bandwidth
GTX	dBi	2.0	3.5	5.0	Antenna gain of a single TX antenna
GRX	dBi	2.0	3.5	5.0	Antenna gain of a single RX antenna
HPBW_RX_E	Deg	50	65	80	Half-power beam width of a single RX antenna in the E-plane direction
HPBW_RX_H	Deg	20	35	50	Half-power beam width of a single RX antenna in the H-plane direction.
HPBW_TX_E	Deg	50	65	80	Half-power beam width of a single TX antenna in the E-plane direction
HPBW_TX_H	Deg	25	40	55	Half-power beam width of a single TX antenna in the H-plane direction
D_RX_RX	mm		2.5		Center-to-center distance between RX antennas in X and Y direction

Abbreviations

6 Abbreviations

Table 31 Abbreviations

Symbol	Description
AAF	Anti-aliasing filter
ADC	Analog to digital converter
AP	Application Processor
DAC	Digital to analog converter
ESD	Electrostatic discharge
FMCW	Frequency modulated continuous wave
HBM	Human body model (related to ESD)
CDM	Charge device model (related to ESD)
HPF	High pass filter
IC	Integrated circuit
LPF	Low pass filter
MCU	Microcontroller Unit
PLL	Phase locked loop integrated circuit
RF	Radio Frequency
RSVD	Reserved
RX	Receiver
SPI	Serial peripheral interface
TX	Transmitter
LDO	Low dropout voltage regulator
RST	Reset or Default setting
MSB	Most significant bit
LSB	Least significant bit

Revision History

7 Revision History

Document version	Date of release	Description of changes
2.4.6	2021-01-08	First Version

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